

Bd. 40 - Gruppen

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# Kapitel 1

## Einleitung

Dieser Band ergänzt die Monoide aus Band 38 um beidseitige Inverse. Er entwickelt die grundlegenden Gruppengesetze, die Eindeutigkeit der Inversen, Kürzung und Gruppenhomomorphismen. Den Abschluss bilden die Einheitengruppe eines Monoids und die Rekonstruktion einer Gruppe aus ihrer großen Potenzhalbgruppe. Abelsche Gruppen werden in Band 41, endliche Gruppen in Band 42 und Ringe in den Bänden 43–44 behandelt. Jede Strukturbezeichnung enthält weiterhin die Operation und ihr neutrales Element, etwa  $\text{Grp}(G, \circ, e)$ .

# Kapitel 2

## Gruppen

### 2.1 Gruppen als Monoide mit Inversen

**Axiom 40.2.1.1** (Existenz eines beidseitigen Inversen).

*Mengen:*  $G, e, x, y$   
*Funktionen:*  $\circ$

$$\text{Mon}(G, \circ, e), x \in G \vdash \\ \exists y \in G (x \circ y = e \wedge y \circ x = e)$$

**Definition 40.2.1.1** (Begriff der Gruppe).

*Mengen:*  $G, e, x, y$   
*Funktionen:*  $\circ$   
*Axiome:*  
*Monoid:*  $\text{Mon}(G, \circ, e)$  Def. 38.2.1.1  
*Beidseitige Inverse:*  $x \in G \vdash$  Ax. 40.2.1.1  
 $\exists y \in G (x \circ y = e \wedge y \circ x = e)$   
*Neues Symbol:*  
*Gruppe:*  $\triangleright \text{Grp}(G, \circ, e)$

$$\text{Grp}(G, \circ, e)$$

**Axiom 40.2.1.2** (Monoidanteil einer Gruppe).

*Mengen:*  $G, e$   
*Funktionen:*  $\circ$

$$\text{Grp}(G, \circ, e) \vdash \text{Mon}(G, \circ, e)$$

**Axiom 40.2.1.3** (Inverse in einer Gruppe).

**Mengen:**  $G, e, x, y$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x \in G \vdash \\ \exists y \in G (x \circ y = e \wedge y \circ x = e)$$

## 2.2 Grundlegende Gruppengesetze

**Theorem 40.2.2.1 (Halbgruppenanteil einer Gruppe).**

**Mengen:**  $G, e$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e) \vdash \text{HGr}(G, \circ)$$

|   |   |                           |                 |
|---|---|---------------------------|-----------------|
| 1 | 1 | $\text{Grp}(G, \circ, e)$ | $A$             |
| 1 | 2 | $\text{Mon}(G, \circ, e)$ | Ax. 40.2.1.2(1) |
| 1 | 3 | $\text{HGr}(G, \circ)$    | Ax. 38.2.1.1(2) |

**Theorem 40.2.2.2 (Abgeschlossenheit der Gruppenoperation).**

**Mengen:**  $G, e, x, y$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x, y \in G \vdash x \circ y \in G$$

|       |   |                           |                       |
|-------|---|---------------------------|-----------------------|
| 1     | 1 | $\text{Grp}(G, \circ, e)$ | $A$                   |
| 2     | 2 | $x \in G$                 | $A$                   |
| 3     | 3 | $y \in G$                 | $A$                   |
| 1     | 4 | $\text{HGr}(G, \circ)$    | Th. 40.2.2.1(1)       |
| 1,2,3 | 5 | $x \circ y \in G$         | Ax. 28.2.1.1(4, 2, 3) |

**Theorem 40.2.2.3 (Assoziativität der Gruppenoperation).**

**Mengen:**  $G, e, x, y, z$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x, y, z \in G \vdash \\ (x \circ y) \circ z = x \circ (y \circ z)$$

|         |   |                                             |                          |
|---------|---|---------------------------------------------|--------------------------|
| 1       | 1 | $\text{Grp}(G, \circ, e)$                   | $A$                      |
| 2       | 2 | $x \in G$                                   | $A$                      |
| 3       | 3 | $y \in G$                                   | $A$                      |
| 4       | 4 | $z \in G$                                   | $A$                      |
| 1       | 5 | $\text{HGr}(G, \circ)$                      | Th. 40.2.2.1(1)          |
| 1,2,3,4 | 6 | $(x \circ y) \circ z = x \circ (y \circ z)$ | Ax. 28.2.1.2(5, 2, 3, 4) |

**Theorem 40.2.2.4 (Neutrales Element liegt im Gruppenträger).**

**Mengen:**  $G, e$

**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e) \vdash e \in G$$

|   |   |                           |                 |
|---|---|---------------------------|-----------------|
| 1 | 1 | $\text{Grp}(G, \circ, e)$ | $A$             |
| 1 | 2 | $\text{Mon}(G, \circ, e)$ | Ax. 40.2.1.2(1) |
| 1 | 3 | $e \in G$                 | Ax. 38.2.1.2(2) |

**Theorem 40.2.2.5 (Linksneutralität in einer Gruppe).**

**Mengen:**  $G, e, x$

**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x \in G \vdash e \circ x = x$$

|     |   |                           |                    |
|-----|---|---------------------------|--------------------|
| 1   | 1 | $\text{Grp}(G, \circ, e)$ | $A$                |
| 2   | 2 | $x \in G$                 | $A$                |
| 1   | 3 | $\text{Mon}(G, \circ, e)$ | Ax. 40.2.1.2(1)    |
| 1,2 | 4 | $e \circ x = x$           | Ax. 38.2.1.3(3, 2) |

**Theorem 40.2.2.6 (Rechtsneutralität in einer Gruppe).**

**Mengen:**  $G, e, x$

**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x \in G \vdash x \circ e = x$$

|     |   |                           |                    |
|-----|---|---------------------------|--------------------|
| 1   | 1 | $\text{Grp}(G, \circ, e)$ | $A$                |
| 2   | 2 | $x \in G$                 | $A$                |
| 1   | 3 | $\text{Mon}(G, \circ, e)$ | Ax. 40.2.1.2(1)    |
| 1,2 | 4 | $x \circ e = x$           | Ax. 38.2.1.4(3, 2) |

## 2.3 Eindeutigkeit des Inversen

**Theorem 40.2.3.1** (Eindeutigkeit aus Links- und Rechtsinverse).

**Mengen:**  $G, e, x, y, z$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), \quad x, y, z \in G, \quad y \circ x = e, \quad x \circ z = e \\ \vdash y = z$$

*Beweis.*

**Einsetzen der Rechtsinverse**

Th. 40.2.3.1(H1)

$$\text{Grp}(G, \circ, e), \quad x, y, z \in G, \quad y \circ x = e, \quad x \circ z = e \\ \vdash y = y \circ (x \circ z)$$

|             |    |                                   |                    |
|-------------|----|-----------------------------------|--------------------|
| 1           | 1  | $\text{Grp}(G, \circ, e)$         | $A$                |
| 2           | 2  | $x \in G$                         | $A$                |
| 3           | 3  | $y \in G$                         | $A$                |
| 4           | 4  | $z \in G$                         | $A$                |
| 5           | 5  | $y \circ x = e$                   | $A$                |
| 6           | 6  | $x \circ z = e$                   | $A$                |
| 1,3         | 7  | $y \circ e = y$                   | Th. 40.2.2.6(1, 3) |
| 1,3         | 8  | $y = y \circ e$                   | Th. 2.9.1.1(7)     |
| 6           | 9  | $y \circ e = y \circ (x \circ z)$ | $= E(6)$           |
| 1,2,3,4,5,6 | 10 | $y = y \circ (x \circ z)$         | Tr.(8, 9)          |

**Reduktion mit der Linksinversen**

Th. 40.2.3.1(H2)

$$\text{Grp}(G, \circ, e), \quad x, y, z \in G, \quad y \circ x = e, \quad x \circ z = e \\ \vdash y \circ (x \circ z) = z$$

|         |   |                                             |                          |
|---------|---|---------------------------------------------|--------------------------|
| 1       | 1 | $\text{Grp}(G, \circ, e)$                   | $A$                      |
| 2       | 2 | $x \in G$                                   | $A$                      |
| 3       | 3 | $y \in G$                                   | $A$                      |
| 4       | 4 | $z \in G$                                   | $A$                      |
| 5       | 5 | $y \circ x = e$                             | $A$                      |
| 6       | 6 | $x \circ z = e$                             | $A$                      |
| 1,2,3,4 | 7 | $(y \circ x) \circ z = y \circ (x \circ z)$ | Th. 40.2.2.3(1, 3, 2, 4) |
| 1,2,3,4 | 8 | $y \circ (x \circ z) = (y \circ x) \circ z$ | Th. 2.9.1.1(7)           |

|                 |    |                                   |                   |
|-----------------|----|-----------------------------------|-------------------|
| 5               | 9  | $(y \circ x) \circ z = e \circ z$ | $= E(5)$          |
| 1,4             | 10 | $e \circ z = z$                   | Th. 40.2.2.5(1,4) |
| 1,2,3,4,5,<br>6 | 11 | $y \circ (x \circ z) = z$         | Tr.(8,9,10)       |

### Verknüpfung beider Teilrechnungen

Th. 40.2.3.1(H3)

$$\text{Grp}(G, \circ, e), \quad x, y, z \in G, \quad y \circ x = e, \quad x \circ z = e \\ \vdash y = z$$

|                 |   |                           |                                   |
|-----------------|---|---------------------------|-----------------------------------|
| 1               | 1 | $\text{Grp}(G, \circ, e)$ | $A$                               |
| 2               | 2 | $x \in G$                 | $A$                               |
| 3               | 3 | $y \in G$                 | $A$                               |
| 4               | 4 | $z \in G$                 | $A$                               |
| 5               | 5 | $y \circ x = e$           | $A$                               |
| 6               | 6 | $x \circ z = e$           | $A$                               |
| 1,2,3,4,5,<br>6 | 7 | $y = y \circ (x \circ z)$ | Th. 40.2.3.1(H1)(1,<br>2,3,4,5,6) |
| 1,2,3,4,5,<br>6 | 8 | $y \circ (x \circ z) = z$ | Th. 40.2.3.1(H2)(1,<br>2,3,4,5,6) |
| 1,2,3,4,5,<br>6 | 9 | $y = z$                   | Tr.(7,8)                          |

□

### Definition 40.2.3.1 (Inverses Element).

|                          |                         |
|--------------------------|-------------------------|
| <b>Mengen:</b>           | $G, e, x, y$            |
| <b>Funktionen:</b>       | $\circ$                 |
| <b>Träger:</b>           | $x^{-1} \in G$          |
| <b>Rechtsinverse:</b>    | $x \circ x^{-1} = e$    |
| <b>Linksinverse:</b>     | $x^{-1} \circ x = e$    |
| <b>Eindeutigkeit:</b>    | Th. 40.2.3.1            |
| <b>Neues Symbol:</b>     |                         |
| <b>Inverses Element:</b> | $\triangleright x^{-1}$ |

$$\text{Grp}(G, \circ, e), \quad x \in G \vdash \\ x^{-1} := \iota y (y \in G \wedge x \circ y = e \wedge y \circ x = e)$$

### Theorem 40.2.3.2 (Inverses Element liegt im Gruppenträger).

|                    |           |
|--------------------|-----------|
| <b>Mengen:</b>     | $G, e, x$ |
| <b>Funktionen:</b> | $\circ$   |

$$\text{Grp}(G, \circ, e), \quad x \in G \vdash x^{-1} \in G$$

|     |   |                           |                     |
|-----|---|---------------------------|---------------------|
| 1   | 1 | $\text{Grp}(G, \circ, e)$ | $A$                 |
| 2   | 2 | $x \in G$                 | $A$                 |
| 1,2 | 3 | $x^{-1} \in G$            | Def. 40.2.3.1(1, 2) |

**Theorem 40.2.3.3 (Rechtsinverse Gleichung).**

**Mengen:**  $G, e, x$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x \in G \vdash x \circ x^{-1} = e$$

|     |   |                           |                     |
|-----|---|---------------------------|---------------------|
| 1   | 1 | $\text{Grp}(G, \circ, e)$ | $A$                 |
| 2   | 2 | $x \in G$                 | $A$                 |
| 1,2 | 3 | $x \circ x^{-1} = e$      | Def. 40.2.3.1(1, 2) |

**Theorem 40.2.3.4 (Linksinverse Gleichung).**

**Mengen:**  $G, e, x$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), x \in G \vdash x^{-1} \circ x = e$$

|     |   |                           |                     |
|-----|---|---------------------------|---------------------|
| 1   | 1 | $\text{Grp}(G, \circ, e)$ | $A$                 |
| 2   | 2 | $x \in G$                 | $A$                 |
| 1,2 | 3 | $x^{-1} \circ x = e$      | Def. 40.2.3.1(1, 2) |

## 2.4 Kürzungsgesetze

**Theorem 40.2.4.1 (Linkskürzung in Gruppen).**

**Mengen:**  $G, e, a, b, c$   
**Funktionen:**  $\circ$

$$\begin{aligned} &\text{Grp}(G, \circ, e), a, b, c \in G, a \circ b = a \circ c \\ &\vdash b = c \end{aligned}$$

*Beweis.*

**Linksmultiplikation mit dem Inversen**

Th. 40.2.4.1(H1)

$$\begin{aligned} &\text{Grp}(G, \circ, e), a, b, c \in G, a \circ b = a \circ c \vdash \\ &(a^{-1} \circ a) \circ b = (a^{-1} \circ a) \circ c \end{aligned}$$

|           |    |                                                       |                          |
|-----------|----|-------------------------------------------------------|--------------------------|
| 1         | 1  | $\text{Grp}(G, \circ, e)$                             | $A$                      |
| 2         | 2  | $a \in G$                                             | $A$                      |
| 3         | 3  | $b \in G$                                             | $A$                      |
| 4         | 4  | $c \in G$                                             | $A$                      |
| 5         | 5  | $a \circ b = a \circ c$                               | $A$                      |
| 1,2       | 6  | $a^{-1} \in G$                                        | Th. 40.2.3.2(1, 2)       |
| 1,2,5     | 7  | $a^{-1} \circ (a \circ b) = a^{-1} \circ (a \circ c)$ | $= E(5)$                 |
| 1,2,3     | 8  | $(a^{-1} \circ a) \circ b = a^{-1} \circ (a \circ b)$ | Th. 40.2.2.3(1, 6, 2, 3) |
| 1,2,4     | 9  | $(a^{-1} \circ a) \circ c = a^{-1} \circ (a \circ c)$ | Th. 40.2.2.3(1, 6, 2, 4) |
| 1,2,4     | 10 | $a^{-1} \circ (a \circ c) = (a^{-1} \circ a) \circ c$ | Th. 2.9.1.1(9)           |
| 1,2,3,4,5 | 11 | $(a^{-1} \circ a) \circ b = (a^{-1} \circ a) \circ c$ | Tr.(8, 7, 10)            |

#### Reduktion auf das neutrale Element

Th. 40.2.4.1(H2)

$$\text{Grp}(G, \circ, e), \quad a, b, c \in G, \quad a \circ b = a \circ c \vdash \\ e \circ b = e \circ c$$

|           |    |                                                       |                                 |
|-----------|----|-------------------------------------------------------|---------------------------------|
| 1         | 1  | $\text{Grp}(G, \circ, e)$                             | $A$                             |
| 2         | 2  | $a \in G$                                             | $A$                             |
| 3         | 3  | $b \in G$                                             | $A$                             |
| 4         | 4  | $c \in G$                                             | $A$                             |
| 5         | 5  | $a \circ b = a \circ c$                               | $A$                             |
| 1,2,3,4,5 | 6  | $(a^{-1} \circ a) \circ b = (a^{-1} \circ a) \circ c$ | Th. 40.2.4.1(H1)(1, 2, 3, 4, 5) |
| 1,2       | 7  | $a^{-1} \circ a = e$                                  | Th. 40.2.3.4(1, 2)              |
| 1,2       | 8  | $e = a^{-1} \circ a$                                  | Th. 2.9.1.1(7)                  |
| 1,2,3     | 9  | $e \circ b = (a^{-1} \circ a) \circ b$                | $= E(8)$                        |
| 1,2,4     | 10 | $(a^{-1} \circ a) \circ c = e \circ c$                | $= E(7)$                        |
| 1,2,3,4,5 | 11 | $e \circ b = e \circ c$                               | Tr.(9, 6, 10)                   |

#### Entfernen des neutralen Elements

Th. 40.2.4.1(H3)

$$\text{Grp}(G, \circ, e), \quad a, b, c \in G, \quad a \circ b = a \circ c \vdash b = c$$

|   |   |                           |     |
|---|---|---------------------------|-----|
| 1 | 1 | $\text{Grp}(G, \circ, e)$ | $A$ |
| 2 | 2 | $a \in G$                 | $A$ |
| 3 | 3 | $b \in G$                 | $A$ |
| 4 | 4 | $c \in G$                 | $A$ |
| 5 | 5 | $a \circ b = a \circ c$   | $A$ |

|           |    |                         |                                 |
|-----------|----|-------------------------|---------------------------------|
| 1,2,3,4,5 | 6  | $e \circ b = e \circ c$ | Th. 40.2.4.1(H2)(1,<br>2,3,4,5) |
| 1,3       | 7  | $e \circ b = b$         | Th. 40.2.2.5(1,3)               |
| 1,3       | 8  | $b = e \circ b$         | Th. 2.9.1.1(7)                  |
| 1,4       | 9  | $e \circ c = c$         | Th. 40.2.2.5(1,4)               |
| 1,2,3,4,5 | 10 | $b = c$                 | Tr.(8,6,9)                      |

□

### Theorem 40.2.4.2 (Rechtskürzung in Gruppen).

**Mengen:**  $G, e, a, b, c$

**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e), a, b, c \in G, b \circ a = c \circ a \\ \vdash b = c$$

*Beweis.*

#### Rechtsmultiplikation mit dem Inversen

Th. 40.2.4.2(H1)

$$\text{Grp}(G, \circ, e), a, b, c \in G, b \circ a = c \circ a \vdash \\ b \circ (a \circ a^{-1}) = c \circ (a \circ a^{-1})$$

|           |    |                                                       |                           |
|-----------|----|-------------------------------------------------------|---------------------------|
| 1         | 1  | $\text{Grp}(G, \circ, e)$                             | A                         |
| 2         | 2  | $a \in G$                                             | A                         |
| 3         | 3  | $b \in G$                                             | A                         |
| 4         | 4  | $c \in G$                                             | A                         |
| 5         | 5  | $b \circ a = c \circ a$                               | A                         |
| 1,2       | 6  | $a^{-1} \in G$                                        | Th. 40.2.3.2(1,2)         |
| 1,2,5     | 7  | $(b \circ a) \circ a^{-1} = (c \circ a) \circ a^{-1}$ | $= E(5)$                  |
| 1,2,3     | 8  | $(b \circ a) \circ a^{-1} = b \circ (a \circ a^{-1})$ | Th. 40.2.2.3(1,3,2,<br>6) |
| 1,2,3     | 9  | $b \circ (a \circ a^{-1}) = (b \circ a) \circ a^{-1}$ | Th. 2.9.1.1(8)            |
| 1,2,4     | 10 | $(c \circ a) \circ a^{-1} = c \circ (a \circ a^{-1})$ | Th. 40.2.2.3(1,4,2,<br>6) |
| 1,2,3,4,5 | 11 | $b \circ (a \circ a^{-1}) = c \circ (a \circ a^{-1})$ | Tr.(9,7,10)               |

#### Reduktion auf das neutrale Element

Th. 40.2.4.2(H2)

$$\text{Grp}(G, \circ, e), a, b, c \in G, b \circ a = c \circ a \vdash \\ b \circ e = c \circ e$$

|           |    |                                                       |                                 |
|-----------|----|-------------------------------------------------------|---------------------------------|
| 1         | 1  | $\text{Grp}(G, \circ, e)$                             | $A$                             |
| 2         | 2  | $a \in G$                                             | $A$                             |
| 3         | 3  | $b \in G$                                             | $A$                             |
| 4         | 4  | $c \in G$                                             | $A$                             |
| 5         | 5  | $b \circ a = c \circ a$                               | $A$                             |
| 1,2,3,4,5 | 6  | $b \circ (a \circ a^{-1}) = c \circ (a \circ a^{-1})$ | Th. 40.2.4.2(H1)(1, 2, 3, 4, 5) |
| 1,2       | 7  | $a \circ a^{-1} = e$                                  | Th. 40.2.3.3(1, 2)              |
| 1,2       | 8  | $e = a \circ a^{-1}$                                  | Th. 2.9.1.1(7)                  |
| 1,2,3     | 9  | $b \circ e = b \circ (a \circ a^{-1})$                | $= E(8)$                        |
| 1,2,4     | 10 | $c \circ (a \circ a^{-1}) = c \circ e$                | $= E(7)$                        |
| 1,2,3,4,5 | 11 | $b \circ e = c \circ e$                               | Tr.(9, 6, 10)                   |

#### Entfernen des neutralen Elements

Th. 40.2.4.2(H3)

$$\text{Grp}(G, \circ, e), a, b, c \in G, b \circ a = c \circ a \vdash b = c$$

|           |    |                           |                                 |
|-----------|----|---------------------------|---------------------------------|
| 1         | 1  | $\text{Grp}(G, \circ, e)$ | $A$                             |
| 2         | 2  | $a \in G$                 | $A$                             |
| 3         | 3  | $b \in G$                 | $A$                             |
| 4         | 4  | $c \in G$                 | $A$                             |
| 5         | 5  | $b \circ a = c \circ a$   | $A$                             |
| 1,2,3,4,5 | 6  | $b \circ e = c \circ e$   | Th. 40.2.4.2(H2)(1, 2, 3, 4, 5) |
| 1,3       | 7  | $b \circ e = b$           | Th. 40.2.2.6(1, 3)              |
| 1,3       | 8  | $b = b \circ e$           | Th. 2.9.1.1(7)                  |
| 1,4       | 9  | $c \circ e = c$           | Th. 40.2.2.6(1, 4)              |
| 1,2,3,4,5 | 10 | $b = c$                   | Tr.(8, 6, 9)                    |

□

**Theorem 40.2.4.3 (Nichtleere endliche kürzbare Halbgruppen sind Gruppen).**

**Mengen:**  $A, a, b, c, d, e, f, x, y$

**Funktionen:**  $\star, L_a, R_a$

$$\begin{aligned} &\text{KürzHGr}(A, \star), \text{Finite}(A), A \neq \emptyset \vdash \\ &\exists e \in A \text{ Grp}(A, \star, e) \end{aligned}$$

|   |   |                            |     |
|---|---|----------------------------|-----|
| 1 | 1 | $\text{KürzHGr}(A, \star)$ | $A$ |
| 2 | 2 | $\text{Finite}(A)$         | $A$ |

|       |   |                                            |                       |
|-------|---|--------------------------------------------|-----------------------|
| 3     | 3 | $A \neq \emptyset$                         | $A$                   |
| 1     | 4 | $\text{HGr}(A, \star)$                     | Ax. 36.2.1.5(1)       |
| 1,2   | 5 | $\text{FinHGr}(A, \star)$                  | Def. 37.2.1.1(4, 2)   |
| 1,2,3 | 6 | $\exists e \in A \text{ Grp}(A, \star, e)$ | Th. 37.5.5.1(5, 1, 3) |

## 2.5 Inverse eines Produkts

**Theorem 40.2.5.1 (Inverse eines Produkts).**

**Mengen:**  $G, e, x, y$

**Funktionen:**  $\circ$

$$\begin{aligned} &\text{Grp}(G, \circ, e), \quad x, y \in G \vdash \\ &(x \circ y)^{-1} = y^{-1} \circ x^{-1} \end{aligned}$$

*Beweis.*

**Rechtsinverse des Produkts**

Th. 40.2.5.1(H1)

$$\begin{aligned} &\text{Grp}(G, \circ, e), \quad x, y \in G \vdash \\ &(x \circ y) \circ (y^{-1} \circ x^{-1}) = e \end{aligned}$$

|       |    |                                                                                     |                               |
|-------|----|-------------------------------------------------------------------------------------|-------------------------------|
| 1     | 1  | $\text{Grp}(G, \circ, e)$                                                           | $A$                           |
| 2     | 2  | $x \in G$                                                                           | $A$                           |
| 3     | 3  | $y \in G$                                                                           | $A$                           |
| 1,2   | 4  | $x^{-1} \in G$                                                                      | Th. 40.2.3.2(1, 2)            |
| 1,3   | 5  | $y^{-1} \in G$                                                                      | Th. 40.2.3.2(1, 3)            |
| 1,2,3 | 6  | $y^{-1} \circ x^{-1} \in G$                                                         | Th. 40.2.2.2(1, 5, 4)         |
| 1,2,3 | 7  | $(x \circ y) \circ (y^{-1} \circ x^{-1}) = x \circ (y \circ (y^{-1} \circ x^{-1}))$ | Th. 40.2.2.3(1, 2, 3, 6)      |
| 1,2,3 | 8  | $(y \circ y^{-1}) \circ x^{-1} = y \circ (y^{-1} \circ x^{-1})$                     | Th. 40.2.2.3(1, 3, 5, 4)      |
| 1,3   | 9  | $y \circ y^{-1} = e$                                                                | Th. 40.2.3.3(1, 3)            |
| 1,2   | 10 | $e \circ x^{-1} = x^{-1}$                                                           | Th. 40.2.2.5(1, 4)            |
| 1,2,3 | 11 | $y \circ (y^{-1} \circ x^{-1}) = x^{-1}$                                            | Th. 2.9.1.4(8, = $E(9, 10)$ ) |
| 1,2,3 | 12 | $x \circ (y \circ (y^{-1} \circ x^{-1})) = x \circ x^{-1}$                          | = $E(11)$                     |
| 1,2   | 13 | $x \circ x^{-1} = e$                                                                | Th. 40.2.3.3(1, 2)            |
| 1,2,3 | 14 | $(x \circ y) \circ (y^{-1} \circ x^{-1}) = e$                                       | Tr.(7, 12, 13)                |

**Linksinverse des Produkts**

Th. 40.2.5.1(H2)

$$\begin{aligned} &\text{Grp}(G, \circ, e), \quad x, y \in G \vdash \\ &(y^{-1} \circ x^{-1}) \circ (x \circ y) = e \end{aligned}$$

|       |    |                                                                                     |                               |
|-------|----|-------------------------------------------------------------------------------------|-------------------------------|
| 1     | 1  | $\text{Grp}(G, \circ, e)$                                                           | $A$                           |
| 2     | 2  | $x \in G$                                                                           | $A$                           |
| 3     | 3  | $y \in G$                                                                           | $A$                           |
| 1,2   | 4  | $x^{-1} \in G$                                                                      | Th. 40.2.3.2(1, 2)            |
| 1,3   | 5  | $y^{-1} \in G$                                                                      | Th. 40.2.3.2(1, 3)            |
| 1,2,3 | 6  | $x \circ y \in G$                                                                   | Th. 40.2.2.2(1, 2, 3)         |
| 1,2,3 | 7  | $(y^{-1} \circ x^{-1}) \circ (x \circ y) = y^{-1} \circ (x^{-1} \circ (x \circ y))$ | Th. 40.2.2.3(1, 5, 4, 6)      |
| 1,2,3 | 8  | $(x^{-1} \circ x) \circ y = x^{-1} \circ (x \circ y)$                               | Th. 40.2.2.3(1, 4, 2, 3)      |
| 1,2   | 9  | $x^{-1} \circ x = e$                                                                | Th. 40.2.3.4(1, 2)            |
| 1,3   | 10 | $e \circ y = y$                                                                     | Th. 40.2.2.5(1, 3)            |
| 1,2,3 | 11 | $x^{-1} \circ (x \circ y) = y$                                                      | Th. 2.9.1.4(8, = $E$ (9, 10)) |
| 1,2,3 | 12 | $y^{-1} \circ (x^{-1} \circ (x \circ y)) = y^{-1} \circ y$                          | = $E(11)$                     |
| 1,3   | 13 | $y^{-1} \circ y = e$                                                                | Th. 40.2.3.4(1, 3)            |
| 1,2,3 | 14 | $(y^{-1} \circ x^{-1}) \circ (x \circ y) = e$                                       | Tr.(7, 12, 13)                |

Eindeutigkeit:

|       |    |                                               |                                 |
|-------|----|-----------------------------------------------|---------------------------------|
| 1     | 1  | $\text{Grp}(G, \circ, e)$                     | $A$                             |
| 2     | 2  | $x \in G$                                     | $A$                             |
| 3     | 3  | $y \in G$                                     | $A$                             |
| 1,2   | 4  | $x^{-1} \in G$                                | Th. 40.2.3.2(1, 2)              |
| 1,3   | 5  | $y^{-1} \in G$                                | Th. 40.2.3.2(1, 3)              |
| 1,2,3 | 6  | $x \circ y \in G$                             | Th. 40.2.2.2(1, 2, 3)           |
| 1,2,3 | 7  | $y^{-1} \circ x^{-1} \in G$                   | Th. 40.2.2.2(1, 5, 4)           |
| 1,2,3 | 8  | $(x \circ y)^{-1} \in G$                      | Th. 40.2.3.2(1, 6)              |
| 1,2,3 | 9  | $(y^{-1} \circ x^{-1}) \circ (x \circ y) = e$ | Th. 40.2.5.1( $H2$ )(1, 2, 3)   |
| 1,2,3 | 10 | $(x \circ y) \circ (x \circ y)^{-1} = e$      | Th. 40.2.3.3(1, 6)              |
| 1,2,3 | 11 | $y^{-1} \circ x^{-1} = (x \circ y)^{-1}$      | Th. 40.2.3.1(1, 6, 7, 8, 9, 10) |
| 1,2,3 | 12 | $(x \circ y)^{-1} = y^{-1} \circ x^{-1}$      | Th. 2.9.1.1(11)                 |

□

## Kapitel 3

# Homomorphismen von Gruppen

### 3.1 Gruppenhomomorphismen

Bei Gruppen genügt die Erhaltung der Gruppenoperation. Anders als bei beliebigen Monoiden folgen die Erhaltung des neutralen Elements und der Inversen aus der Existenz von Inversen im Ziel.

**Definition 40.3.1.1 (Begriff des Gruppenhomomorphismus).**

|                                   |                                                                |
|-----------------------------------|----------------------------------------------------------------|
| <i>Mengen:</i>                    | $G, H, e, u, x, y$                                             |
| <i>Funktionen:</i>                | $f, \circ, \diamond$                                           |
| <i>Axiome:</i>                    |                                                                |
| <i>Quellgruppe:</i>               | $\triangleright \text{Grp}(G, \circ, e)$                       |
| <i>Zielgruppe:</i>                | $\triangleright \text{Grp}(H, \diamond, u)$                    |
| <i>Halbgruppenhomomorphismus:</i> | $\triangleright \text{HGrHom}(f; G, \circ; H, \diamond)$       |
| <i>Neues Symbol:</i>              |                                                                |
| <i>Gruppenhomomorphismus:</i>     | $\triangleright \text{GrpHom}(f; G, \circ, e; H, \diamond, u)$ |

$$\text{GrpHom}(f; G, \circ, e; H, \diamond, u)$$

**Axiom 40.3.1.1 (Trägerabbildung eines Gruppenhomomorphismus).**

|                    |                      |
|--------------------|----------------------|
| <i>Mengen:</i>     | $G, H, e, u$         |
| <i>Funktionen:</i> | $f, \circ, \diamond$ |

$$\text{GrpHom}(f; G, \circ, e; H, \diamond, u) \vdash f: G \rightarrow H$$

**Axiom 40.3.1.2 (Operationserhaltung eines Gruppenhomomorphismus).**

|                    |                      |
|--------------------|----------------------|
| <i>Mengen:</i>     | $G, H, e, u, x, y$   |
| <i>Funktionen:</i> | $f, \circ, \diamond$ |

$$\text{GrpHom}(f; G, \circ, e; H, \diamond, u), x, y \in G \vdash f(x \circ y) = f(x) \diamond f(y)$$

**Theorem 40.3.1.1 (Gruppenhomomorphismen erhalten das neutrale Element).**

**Mengen:**  $G, H, e, u$

**Funktionen:**  $f, \circ, \diamond$

$$\text{GrpHom}(f; G, \circ, e; H, \diamond, u) \vdash f(e) = u$$

|   |    |                                                 |                                    |
|---|----|-------------------------------------------------|------------------------------------|
| 1 | 1  | $\text{GrpHom}(f; G, \circ, e; H, \diamond, u)$ | $A$                                |
| 1 | 2  | $\text{Grp}(G, \circ, e)$                       | Def. 40.3.1.1(1)                   |
| 1 | 3  | $\text{Grp}(H, \diamond, u)$                    | Def. 40.3.1.1(1)                   |
| 1 | 4  | $f: G \rightarrow H$                            | Ax. 40.3.1.1(1)                    |
| 1 | 5  | $e \in G$                                       | Th. 40.2.2.4(2)                    |
| 1 | 6  | $f(e) \in H$                                    | Th. 5.2.3.9(4, 5)                  |
| 1 | 7  | $e \circ e = e$                                 | Th. 40.2.2.5(2, 5)                 |
| 1 | 8  | $f(e \circ e) = f(e)$                           | $= E(7)$                           |
| 1 | 9  | $f(e \circ e) = f(e) \diamond f(e)$             | Ax. 40.3.1.2(1, 5, 5)              |
| 1 | 10 | $f(e) \diamond f(e) = f(e)$                     | $= E(9, 8)$                        |
| 1 | 11 | $f(e) \diamond u = f(e)$                        | Th. 40.2.2.6(3, 6)                 |
| 1 | 12 | $f(e) \diamond u = f(e) \diamond f(e)$          | $= E(11, \text{Th. 2.9.1.1 (10)})$ |
| 1 | 13 | $u \in H$                                       | Th. 40.2.2.4(3)                    |
| 1 | 14 | $u = f(e)$                                      | Th. 40.2.4.1(3, 6, 13, 6, 12)      |
| 1 | 15 | $f(e) = u$                                      | Th. 2.9.1.1(14)                    |

**Theorem 40.3.1.2 (Gruppenhomomorphismen erhalten Inverse).**

**Mengen:**  $G, H, e, u, x$

**Funktionen:**  $f, \circ, \diamond$

$$\text{GrpHom}(f; G, \circ, e; H, \diamond, u), x \in G \vdash f(x^{-1}) = f(x)^{-1}$$

*Beweis.*

Träger der Bilder:

|   |   |                                                 |                  |
|---|---|-------------------------------------------------|------------------|
| 1 | 1 | $\text{GrpHom}(f; G, \circ, e; H, \diamond, u)$ | $A$              |
| 2 | 2 | $x \in G$                                       | $A$              |
| 1 | 3 | $\text{Grp}(G, \circ, e)$                       | Def. 40.3.1.1(1) |

|     |   |                              |                    |
|-----|---|------------------------------|--------------------|
| 1   | 4 | $\text{Grp}(H, \diamond, u)$ | Def. 40.3.1.1(1)   |
| 1   | 5 | $f: G \rightarrow H$         | Ax. 40.3.1.1(1)    |
| 1,2 | 6 | $x^{-1} \in G$               | Th. 40.2.3.2(3, 2) |
| 1,2 | 7 | $f(x) \in H$                 | Th. 5.2.3.9(5, 2)  |
| 1,2 | 8 | $f(x^{-1}) \in H$            | Th. 5.2.3.9(5, 6)  |

Linksinverse und Eindeutigkeit:

|     |    |                                               |                                   |
|-----|----|-----------------------------------------------|-----------------------------------|
| 1   | 9  | $f(e) = u$                                    | Th. 40.3.1.1(1)                   |
| 1,2 | 10 | $x^{-1} \circ x = e$                          | Th. 40.2.3.4(3, 2)                |
| 1,2 | 11 | $f(x^{-1} \circ x) = f(x^{-1}) \diamond f(x)$ | Ax. 40.3.1.2(1, 6, 2)             |
| 1,2 | 12 | $f(x^{-1} \circ x) = u$                       | $= E(10, 9)$                      |
| 1,2 | 13 | $f(x^{-1}) \diamond f(x) = u$                 | Th. 2.9.1.4(11, 12)               |
| 1,2 | 14 | $f(x)^{-1} \in H$                             | Th. 40.2.3.2(4, 7)                |
| 1,2 | 15 | $f(x) \diamond f(x)^{-1} = u$                 | Th. 40.2.3.3(4, 7)                |
| 1,2 | 16 | $f(x^{-1}) = f(x)^{-1}$                       | Th. 40.2.3.1(4, 7, 8, 14, 13, 15) |

□

**Theorem 40.3.1.3 (Ein Gruppenhomomorphismus ist ein Monoidhomomorphismus).**

**Mengen:**  $G, H, e, u$   
**Funktionen:**  $f, \circ, \diamond$

$$\text{GrpHom}(f; G, \circ, e; H, \diamond, u) \vdash \text{MonHom}(f; G, \circ, e; H, \diamond, u)$$

|   |   |                                                 |                           |
|---|---|-------------------------------------------------|---------------------------|
| 1 | 1 | $\text{GrpHom}(f; G, \circ, e; H, \diamond, u)$ | $A$                       |
| 1 | 2 | $\text{Grp}(G, \circ, e)$                       | Def. 40.3.1.1(1)          |
| 1 | 3 | $\text{Grp}(H, \diamond, u)$                    | Def. 40.3.1.1(1)          |
| 1 | 4 | $\text{Mon}(G, \circ, e)$                       | Ax. 40.2.1.2(2)           |
| 1 | 5 | $\text{Mon}(H, \diamond, u)$                    | Ax. 40.2.1.2(3)           |
| 1 | 6 | $\text{HGrHom}(f; G, \circ; H, \diamond)$       | Def. 40.3.1.1(1)          |
| 1 | 7 | $f(e) = u$                                      | Th. 40.3.1.1(1)           |
| 1 | 8 | $\text{MonHom}(f; G, \circ, e; H, \diamond, u)$ | Def. 38.2.6.1(4, 5, 6, 7) |

## 3.2 Gruppenisomorphismen und Komposition

**Definition 40.3.2.1 (Begriff des Gruppenisomorphismus).**

**Mengen:**  $G, H, e, u$   
**Funktionen:**  $f, \circ, \diamond$   
**Axiome:**  
**Gruppenhomomorphismus:**  $\triangleright \text{GrpHom}(f; G, \circ, e; H, \diamond, u)$   
**Bijektivität:**  $\triangleright f: G \xrightarrow{\sim} H$   
**Neues Symbol:**  
**Gruppenisomorphismus:**  $\triangleright \text{GrpIso}(f; G, \circ, e; H, \diamond, u)$

$$\text{GrpIso}(f; G, \circ, e; H, \diamond, u)$$

**Theorem 40.3.2.1 (Komposition von Gruppenhomomorphismen).**

**Mengen:**  $G, H, K, e, u, v$   
**Funktionen:**  $f, g, \circ, \diamond, \bullet$

$$\begin{aligned} & \text{GrpHom}(f; G, \circ, e; H, \diamond, u), \text{GrpHom}(g; H, \diamond, u; K, \bullet, v) \\ & \vdash \text{GrpHom}(g \circ f; G, \circ, e; K, \bullet, v) \end{aligned}$$

|     |   |                                                        |                        |
|-----|---|--------------------------------------------------------|------------------------|
| 1   | 1 | $\text{GrpHom}(f; G, \circ, e; H, \diamond, u)$        | $A$                    |
| 2   | 2 | $\text{GrpHom}(g; H, \diamond, u; K, \bullet, v)$      | $A$                    |
| 1   | 3 | $\text{HGrHom}(f; G, \circ; H, \diamond)$              | Def. 40.3.1.1(1)       |
| 2   | 4 | $\text{HGrHom}(g; H, \diamond; K, \bullet)$            | Def. 40.3.1.1(2)       |
| 1,2 | 5 | $\text{HGrHom}(g \circ f; G, \circ; K, \bullet)$       | Th. 28.2.12.1(3, 4)    |
| 1   | 6 | $\text{Grp}(G, \circ, e)$                              | Def. 40.3.1.1(1)       |
| 2   | 7 | $\text{Grp}(K, \bullet, v)$                            | Def. 40.3.1.1(2)       |
| 1,2 | 8 | $\text{GrpHom}(g \circ f; G, \circ, e; K, \bullet, v)$ | Def. 40.3.1.1(6, 7, 5) |

**Theorem 40.3.2.2 (Die Identität ist ein Gruppenisomorphismus).**

**Mengen:**  $G, e$   
**Funktionen:**  $\circ$

$$\text{Grp}(G, \circ, e) \vdash \text{GrpIso}(\text{id}_G; G, \circ, e; G, \circ, e)$$

|   |   |                                                        |                        |
|---|---|--------------------------------------------------------|------------------------|
| 1 | 1 | $\text{Grp}(G, \circ, e)$                              | $A$                    |
| 1 | 2 | $\text{HGr}(G, \circ)$                                 | Th. 40.2.2.1(1)        |
| 1 | 3 | $\text{HGrIso}(\text{id}_G; G, \circ; G, \circ)$       | Th. 28.2.12.2(2)       |
| 1 | 4 | $\text{HGrHom}(\text{id}_G; G, \circ; G, \circ)$       | Ax. 28.2.12.5(3)       |
| 1 | 5 | $\text{GrpHom}(\text{id}_G; G, \circ, e; G, \circ, e)$ | Def. 40.3.1.1(1, 1, 4) |
|   | 6 | $\text{id}_G: G \xrightarrow{\sim} G$                  | Th. 8.3.1.7            |
| 1 | 7 | $\text{GrpIso}(\text{id}_G; G, \circ, e; G, \circ, e)$ | Def. 40.3.2.1(5, 6)    |

### 3.3 Rekonstruktion aus der großen Potenzhalbgruppe

#### 3.3.1 Der konkrete Transport von Gruppeneinermengen

Für zwei Gruppen halten wir fest, welche Einermenge ein gegebener Potenzisomorphismus wohin schickt. Diese Aussage wird anschließend für die einseitige Gruppenstarrheit und später für Faserargumente verwendet.

**Theorem 40.3.3.1 (Jedes Gruppenelement ist eine Einheit).**

**Mengen:**  $G, e, g, h$

**Funktionen:**  $\star$

$$\text{Grp}(G, \star, e), g \in G \vdash \text{Einh}_{G, \star, e}(g)$$

|     |   |                                                        |                        |
|-----|---|--------------------------------------------------------|------------------------|
| 1   | 1 | $\text{Grp}(G, \star, e)$                              | $A$                    |
| 2   | 2 | $g \in G$                                              | $A$                    |
| 1   | 3 | $\text{Mon}(G, \star, e)$                              | Ax. 40.2.1.2(1)        |
| 1,2 | 4 | $\exists h \in G (g \star h = e \wedge h \star g = e)$ | Ax. 40.2.1.3(1, 2)     |
| 1,2 | 5 | $\text{Einh}_{G, \star, e}(g)$                         | Def. 38.2.5.1(3, 2, 4) |

**Theorem 40.3.3.2 (Einheiten einer Gruppenpotenzhalbgruppe sind genau Einermengen).**

**Mengen:**  $G, e, X, g$

**Funktionen:**  $\star, \star_P$

$$\text{Grp}(G, \star, e), X \in \mathcal{P}(G) \setminus \{\emptyset\} \vdash \\ \text{Einh}_{\mathcal{P}(G) \setminus \{\emptyset\}, \star_P, \{e\}}(X) \leftrightarrow X \in \text{Sing}(G)$$

|     |    |                                                                                                                                                                |                             |
|-----|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 1   | 1  | $\text{Grp}(G, \star, e)$                                                                                                                                      | $A$                         |
| 2   | 2  | $X \in \mathcal{P}(G) \setminus \{\emptyset\}$                                                                                                                 | $A$                         |
| 1   | 3  | $\text{Mon}(G, \star, e)$                                                                                                                                      | Ax. 40.2.1.2(1)             |
| 1,2 | 4  | $\text{Einh}_{\mathcal{P}(G) \setminus \{\emptyset\}, \star_P, \{e\}}(X)$<br>$\leftrightarrow \exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$ | Th. 38.2.5.1(3, 2)          |
| 5   | 5  | $\exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$                                                                                              | $A$                         |
| 6   | 6  | $g \in G \wedge (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$                                                                                               | $A$                         |
| 6   | 7  | $g \in G$                                                                                                                                                      | $\wedge E1(6)$              |
| 6   | 8  | $X = \{g\}$                                                                                                                                                    | $\wedge E1(\wedge E2(6))$   |
| 6   | 9  | $\exists g \in G (X = \{g\})$                                                                                                                                  | $\exists I(\wedge I(7, 8))$ |
| 5   | 10 | $\exists g \in G (X = \{g\})$                                                                                                                                  | $\exists E(5, 6, 9)$        |

|      |                                                                                       |                                                   |
|------|---------------------------------------------------------------------------------------|---------------------------------------------------|
| 11   | $\exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$                     | $\rightarrow I(5, 10)$                            |
|      | $\rightarrow \exists g \in G (X = \{g\})$                                             |                                                   |
| 12   | $\exists g \in G (X = \{g\})$                                                         | $A$                                               |
| 13   | $g \in G \wedge X = \{g\}$                                                            | $A$                                               |
| 13   | $g \in G$                                                                             | $\wedge E1(13)$                                   |
| 13   | $X = \{g\}$                                                                           | $\wedge E2(13)$                                   |
| 1,13 | $\text{Einh}_{G, \star, e}(g)$                                                        | Th. 40.3.3.1(1, 14)                               |
| 1,13 | $\exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$                     | $\exists I(\wedge I(14, \wedge I(15,$<br>$16)))$  |
| 1,12 | $\exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$                     | $\exists E(12, 13, 17)$                           |
| 1    | $\exists g \in G (X = \{g\})$                                                         | $\rightarrow I(12, 18)$                           |
|      | $\rightarrow \exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$         |                                                   |
| 1    | $\exists g \in G (X = \{g\} \wedge \text{Einh}_{G, \star, e}(g))$                     | $\leftrightarrow I(11, 19)$                       |
|      | $\leftrightarrow \exists g \in G (X = \{g\})$                                         |                                                   |
| 1    | $\text{HGr}(G, \star)$                                                                | Th. 40.2.2.1(1)                                   |
| 1    | $X \in \text{Sing}(G) \leftrightarrow \exists g \in G (X = \{g\})$                    | Def. 28.2.4.2(21)                                 |
| 1,2  | $\text{Einh}_{\mathcal{P}(G) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\}}(X)$ | $\leftrightarrow S(22, \leftrightarrow S(20, 4))$ |
|      | $\leftrightarrow X \in \text{Sing}(G)$                                                |                                                   |

**Theorem 40.3.3.3 (Potenzisomorphismen zwischen Gruppen erhalten die Einermengenschicht).**

**Mengen:**  $G, H, e, u, X$

**Funktionen:**  $F, R = F \upharpoonright_{\text{Sing}(G)}, \star, \diamond$

$$\begin{aligned} & \text{Grp}(G, \star, e), \text{Grp}(H, \diamond, u), \\ & \text{HGrIso}(F; \mathcal{P}(G) \setminus \{\emptyset\}, \star_{\mathcal{P}}; \mathcal{P}(H) \setminus \{\emptyset\}, \diamond_{\mathcal{P}}) \vdash \\ & F[\text{Sing}(G)] = \text{Sing}(H) \end{aligned}$$

|         |                                                                                                                                                 |                       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1       | $\text{Grp}(G, \star, e)$                                                                                                                       | $A$                   |
| 2       | $\text{Grp}(H, \diamond, u)$                                                                                                                    | $A$                   |
| 3       | $\text{HGrIso}(F; \mathcal{P}(G) \setminus \{\emptyset\}, \star_{\mathcal{P}}; \mathcal{P}(H) \setminus \{\emptyset\}, \diamond_{\mathcal{P}})$ | $A$                   |
| 1       | $\text{Mon}(G, \star, e)$                                                                                                                       | Ax. 40.2.1.2(1)       |
| 1       | $\text{Mon}(\mathcal{P}(G) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\})$                                                                | Th. 38.2.2.3(4)       |
| 2       | $\text{Mon}(H, \diamond, u)$                                                                                                                    | Ax. 40.2.1.2(2)       |
| 2       | $\text{Mon}(\mathcal{P}(H) \setminus \{\emptyset\}, \diamond_{\mathcal{P}}, \{u\})$                                                             | Th. 38.2.2.3(6)       |
| 3       | $F: \mathcal{P}(G) \setminus \{\emptyset\} \xrightarrow{\sim} \mathcal{P}(H) \setminus \{\emptyset\}$                                           | Ax. 28.2.12.6(3)      |
| 9       | $X \in \mathcal{P}(G) \setminus \{\emptyset\}$                                                                                                  | $A$                   |
| 3,9     | $F(X) \in \mathcal{P}(H) \setminus \{\emptyset\}$                                                                                               | Th. 8.2.1.2(8, 9)     |
| 1,2,3,9 | $\text{Einh}_{\mathcal{P}(G) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\}}(X)$                                                           | Th. 38.2.6.5(5, 7, 3, |
|         | $\leftrightarrow \text{Einh}_{\mathcal{P}(H) \setminus \{\emptyset\}, \diamond_{\mathcal{P}}, \{u\}}(F(X))$                                     | 9)                    |

|         |    |                                                                                                                                                                    |                                                                                                   |
|---------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1,2,3,9 | 12 | $X \in \text{Sing}(G) \leftrightarrow F(X) \in \text{Sing}(H)$                                                                                                     | $\leftrightarrow S(\text{Th. 40.3.3.2}(2, 10), \leftrightarrow S(\text{Th. 40.3.3.2}(1, 9), 11))$ |
| 1,2,3   | 13 | $\forall X \in \mathcal{P}(G) \setminus \{\emptyset\} \left( \begin{array}{l} X \in \text{Sing}(G) \\ \leftrightarrow F(X) \in \text{Sing}(H) \end{array} \right)$ | $\forall I(\rightarrow I(9, 12))$                                                                 |
| 1       | 14 | $\text{HGr}(G, \star)$                                                                                                                                             | Th. 40.2.2.1(1)                                                                                   |
| 1       | 15 | $\text{Sing}(G) \subseteq \mathcal{P}(G) \setminus \{\emptyset\}$                                                                                                  | Th. 28.2.14.17(14)                                                                                |
| 2       | 16 | $\text{HGr}(H, \diamond)$                                                                                                                                          | Th. 40.2.2.1(2)                                                                                   |
| 2       | 17 | $\text{Sing}(H) \subseteq \mathcal{P}(H) \setminus \{\emptyset\}$                                                                                                  | Th. 28.2.14.17(16)                                                                                |
| 1,2,3   | 18 | $R: \text{Sing}(G) \xrightarrow{\sim} \text{Sing}(H)$                                                                                                              | Th. 8.3.2.2( $H4$ )(8, 15, 17, 13)                                                                |
| 1,2,3   | 19 | $R[\text{Sing}(G)] = \text{Sing}(H)$                                                                                                                               | Th. 8.2.2.1(18)                                                                                   |
| 1,3     | 20 | $F: \mathcal{P}(G) \setminus \{\emptyset\} \rightarrow \mathcal{P}(H) \setminus \{\emptyset\}$                                                                     | Def. 8.2.1.1(8)                                                                                   |
| 1,3     | 21 | $R[\text{Sing}(G)] = F[\text{Sing}(G)]$                                                                                                                            | Th. 5.3.2.16(15, 20)                                                                              |
| 1,2,3   | 22 | $F[\text{Sing}(G)] = \text{Sing}(H)$                                                                                                                               | Th. 2.9.1.4(21, 19)                                                                               |

**Theorem 40.3.3.4 (Gruppenrekonstruktion mit vorgegebenen Eismengenbildern).**

**Mengen:**  $G, H, e, u, g$

**Funktionen:**  $F, \varphi, \star, \diamond$

$$\begin{aligned} & \text{Grp}(G, \star, e), \text{Grp}(H, \diamond, u), \\ & \text{HGrIso}(F; \mathcal{P}(G) \setminus \{\emptyset\}, \star_P; \mathcal{P}(H) \setminus \{\emptyset\}, \diamond_P) \vdash \\ & \exists \varphi \left( \begin{array}{l} \text{GrpIso}(\varphi; G, \star, e; H, \diamond, u) \\ \wedge \forall g \in G (F(\{g\}) = \{\varphi(g)\}) \end{array} \right) \end{aligned}$$

|       |   |                                                                                                                                                                   |                                 |
|-------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1     | 1 | $\text{Grp}(G, \star, e)$                                                                                                                                         | $A$                             |
| 2     | 2 | $\text{Grp}(H, \diamond, u)$                                                                                                                                      | $A$                             |
| 3     | 3 | $\text{HGrIso}(F; \mathcal{P}(G) \setminus \{\emptyset\}, \star_P; \mathcal{P}(H) \setminus \{\emptyset\}, \diamond_P)$                                           | $A$                             |
| 1,2,3 | 4 | $F[\text{Sing}(G)] = \text{Sing}(H)$                                                                                                                              | Th. 40.3.3.3(1, 2, 3)           |
| 1,2,3 | 5 | $\text{HGr}(G, \star)$                                                                                                                                            | Th. 40.2.2.1(1)                 |
| 1,2,3 | 6 | $\text{HGr}(H, \diamond)$                                                                                                                                         | Th. 40.2.2.1(2)                 |
| 1,2,3 | 7 | $\exists \varphi \left( \begin{array}{l} \text{HGrIso}(\varphi; G, \star; H, \diamond) \\ \wedge \forall g \in G (F(\{g\}) = \{\varphi(g)\}) \end{array} \right)$ | Th. 28.2.14.18(5, 6, 3, 4)      |
| 8     | 8 | $\text{HGrIso}(\varphi; G, \star; H, \diamond)$                                                                                                                   | $A$                             |
|       |   | $\wedge \forall g \in G (F(\{g\}) = \{\varphi(g)\})$                                                                                                              |                                 |
| 8     | 9 | $\text{HGrHom}(\varphi; G, \star; H, \diamond)$                                                                                                                   | Ax. 28.2.12.5( $\wedge E1$ (8)) |

|       |    |                                                                                                                                                                         |                                         |
|-------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 8     | 10 | $\varphi: G \xrightarrow{\sim} H$                                                                                                                                       | Ax. 28.2.12.6( $\wedge E1$<br>(8))      |
| 1,2,8 | 11 | $\text{GrpHom}(\varphi; G, \star, e; H, \diamond, u)$                                                                                                                   | Def. 40.3.1.1(1, 2, 9)                  |
| 1,2,8 | 12 | $\text{Grplso}(\varphi; G, \star, e; H, \diamond, u)$                                                                                                                   | Def. 40.3.2.1(11, 10)                   |
| 1,2,8 | 13 | $\exists \varphi \left( \begin{array}{l} \text{Grplso}(\varphi; G, \star, e; H, \diamond, u) \\ \wedge \forall g \in G (F(\{g\}) = \{\varphi(g)\}) \end{array} \right)$ | $\exists I(\wedge I(12, \wedge E2(8)))$ |
| 1,2,3 | 14 | $\exists \varphi \left( \begin{array}{l} \text{Grplso}(\varphi; G, \star, e; H, \diamond, u) \\ \wedge \forall g \in G (F(\{g\}) = \{\varphi(g)\}) \end{array} \right)$ | $\exists E(7, 8, 13)$                   |

### 3.3.2 Einheitenmengen und Einheitenstabilität

**Definition 40.3.3.1 (Einheitenmenge eines Monoids).**

**Mengen:**  $A, e, u$   
**Funktionen:**  $\star$

$$\text{Mon}(A, \star, e) \vdash A^\times := \{u \in A \mid \text{Einh}_{A, \star, e}(u)\}$$

**Theorem 40.3.3.5 (Inverse bleiben in der Einheitenmenge).**

**Mengen:**  $A, e, u, v$   
**Funktionen:**  $\star$

$$\text{Mon}(A, \star, e), u \in A^\times \vdash \exists v \in A^\times (u \star v = e \wedge v \star u = e)$$

|       |    |                                                               |                                                                                                  |
|-------|----|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1     | 1  | $\text{Mon}(A, \star, e)$                                     | $A$                                                                                              |
| 2     | 2  | $u \in A^\times$                                              | $A$                                                                                              |
| 1,2   | 3  | $u \in A$                                                     | Def. 40.3.3.1(1, 2)                                                                              |
| 1,2   | 4  | $\text{Einh}_{A, \star, e}(u)$                                | Def. 40.3.3.1(1, 2)                                                                              |
| 1,2   | 5  | $\exists v \in A (u \star v = e \wedge v \star u = e)$        | Def. 38.2.5.1(4)                                                                                 |
| 6     | 6  | $v \in A \wedge (u \star v = e \wedge v \star u = e)$         | $A$                                                                                              |
| 6     | 7  | $v \in A$                                                     | $\wedge E1(6)$                                                                                   |
| 6     | 8  | $u \star v = e \wedge v \star u = e$                          | $\wedge E2(6)$                                                                                   |
| 1,2,6 | 9  | $\text{Einh}_{A, \star, e}(v)$                                | Def. 38.2.5.1(1, 7, $\exists I$<br>( $\wedge I(3, \wedge I(\wedge E2(8),$<br>$\wedge E1(8))))$ ) |
| 1,2,6 | 10 | $v \in A^\times$                                              | Def. 40.3.3.1(1, 7, 9)                                                                           |
| 1,2,6 | 11 | $\exists v \in A^\times (u \star v = e \wedge v \star u = e)$ | $\exists I(\wedge I(10, 8))$                                                                     |
| 1,2   | 12 | $\exists v \in A^\times (u \star v = e \wedge v \star u = e)$ | $\exists E(5, 6, 11)$                                                                            |

**Theorem 40.3.3.6 (Inverse eines Produkts im Monoid).**

**Mengen:**  $A, e, u, v, u', v'$   
**Funktionen:**  $\star$

$$\begin{aligned} & \text{Mon}(A, \star, e), \quad u, v, u', v' \in A, \\ & u \star u' = e, \quad u' \star u = e, \quad v \star v' = e, \quad v' \star v = e \vdash \\ & (u \star v) \star (v' \star u') = e \wedge (v' \star u') \star (u \star v) = e \end{aligned}$$

*Beweis.*

Voraussetzungen:

|   |    |                           |                 |
|---|----|---------------------------|-----------------|
| 1 | 1  | $\text{Mon}(A, \star, e)$ | $A$             |
| 2 | 2  | $u \in A$                 | $A$             |
| 3 | 3  | $v \in A$                 | $A$             |
| 4 | 4  | $u' \in A$                | $A$             |
| 5 | 5  | $v' \in A$                | $A$             |
| 6 | 6  | $u \star u' = e$          | $A$             |
| 7 | 7  | $u' \star u = e$          | $A$             |
| 8 | 8  | $v \star v' = e$          | $A$             |
| 9 | 9  | $v' \star v = e$          | $A$             |
| 1 | 10 | $\text{HGr}(A, \star)$    | Ax. 38.2.1.1(1) |

Rechtsinverse des Produkts:

|                |    |                                                                     |                            |
|----------------|----|---------------------------------------------------------------------|----------------------------|
| 1,4,5          | 11 | $v' \star u' \in A$                                                 | Ax. 28.2.1.1(10, 5, 4)     |
| 1,2,3,4,5      | 12 | $(u \star v) \star (v' \star u') = u \star (v \star (v' \star u'))$ | Ax. 28.2.1.2(10, 2, 3, 11) |
| 1,3,4,5        | 13 | $(v \star v') \star u' = v \star (v' \star u')$                     | Ax. 28.2.1.2(10, 3, 5, 4)  |
| 1,3,4,5,8      | 14 | $u \star (v \star (v' \star u')) = u \star (e \star u')$            | $= E(13, 8)$               |
| 1,4            | 15 | $e \star u' = u'$                                                   | Ax. 38.2.1.3(1, 4)         |
| 1,4,6          | 16 | $u \star (e \star u') = e$                                          | $= E(15, 6)$               |
| 1,2,3,4,5, 6,8 | 17 | $(u \star v) \star (v' \star u') = e$                               | Tr.(12, 14, 16)            |

Linksinverse des Produkts:

|           |    |                                                                     |                            |
|-----------|----|---------------------------------------------------------------------|----------------------------|
| 1,2,3     | 18 | $u \star v \in A$                                                   | Ax. 28.2.1.1(10, 2, 3)     |
| 1,2,3,4,5 | 19 | $(v' \star u') \star (u \star v) = v' \star (u' \star (u \star v))$ | Ax. 28.2.1.2(10, 5, 4, 18) |
| 1,2,3,4   | 20 | $(u' \star u) \star v = u' \star (u \star v)$                       | Ax. 28.2.1.2(10, 4, 2, 3)  |
| 1,2,3,4,7 | 21 | $v' \star (u' \star (u \star v)) = v' \star (e \star v)$            | $= E(20, 7)$               |
| 1,3       | 22 | $e \star v = v$                                                     | Ax. 38.2.1.3(1, 3)         |
| 1,3,9     | 23 | $v' \star (e \star v) = e$                                          | $= E(22, 9)$               |

|                       |    |                                                                                  |                    |
|-----------------------|----|----------------------------------------------------------------------------------|--------------------|
| 1,2,3,4,5,<br>7,9     | 24 | $(v' \star u') \star (u \star v) = e$                                            | Tr.(19, 21, 23)    |
| 1,2,3,4,5,<br>6,7,8,9 | 25 | $(u \star v) \star (v' \star u') = e \wedge (v' \star u') \star (u \star v) = e$ | $\wedge I(17, 24)$ |

□

**Theorem 40.3.3.7 (Produktabschluss der Einheitenmenge).**

**Mengen:**  $A, e, u, v, u', v'$

**Funktionen:**  $\star$

$$\text{Mon}(A, \star, e), u, v \in A^\times \vdash u \star v \in A^\times$$

*Beweis.*

Inverse der Faktoren:

|     |    |                                                                  |                                      |
|-----|----|------------------------------------------------------------------|--------------------------------------|
| 1   | 1  | $\text{Mon}(A, \star, e)$                                        | $A$                                  |
| 2   | 2  | $u \in A^\times$                                                 | $A$                                  |
| 3   | 3  | $v \in A^\times$                                                 | $A$                                  |
| 1,2 | 4  | $u \in A$                                                        | Def. 40.3.3.1(1, 2)                  |
| 1,2 | 5  | $\exists u' \in A^\times (u \star u' = e \wedge u' \star u = e)$ | Th. 40.3.3.5(1, 2)                   |
| 1,3 | 6  | $v \in A$                                                        | Def. 40.3.3.1(1, 3)                  |
| 1,3 | 7  | $\exists v' \in A^\times (v \star v' = e \wedge v' \star v = e)$ | Th. 40.3.3.5(1, 3)                   |
| 8   | 8  | $u' \in A^\times \wedge (u \star u' = e \wedge u' \star u = e)$  | $A$                                  |
| 9   | 9  | $v' \in A^\times \wedge (v \star v' = e \wedge v' \star v = e)$  | $A$                                  |
| 1,8 | 10 | $u' \in A$                                                       | Def. 40.3.3.1(1, $\wedge E1$<br>(8)) |
| 1,9 | 11 | $v' \in A$                                                       | Def. 40.3.3.1(1, $\wedge E1$<br>(9)) |

Produktinverse und Abschluss:

|           |    |                                                                                  |                                                                                                                                                               |
|-----------|----|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | 12 | $\text{HGr}(A, \star)$                                                           | Ax. 38.2.1.1(1)                                                                                                                                               |
| 1,2,3     | 13 | $u \star v \in A$                                                                | Ax. 28.2.1.1(12, 4, 6)                                                                                                                                        |
| 1,8,9     | 14 | $v' \star u' \in A$                                                              | Ax. 28.2.1.1(12, 11,<br>10)                                                                                                                                   |
| 1,2,3,8,9 | 15 | $(u \star v) \star (v' \star u') = e \wedge (v' \star u') \star (u \star v) = e$ | Th. 40.3.3.6(1, 4, 6,<br>10, 11, $\wedge E1(\wedge E2$<br>(8)), $\wedge E2(\wedge E2(8))$ ,<br>$\wedge E1(\wedge E2(9))$ , $\wedge E2$<br>( $\wedge E2(9)$ )) |
| 1,2,3,8,9 | 16 | $\text{Einh}_{A, \star, e}(u \star v)$                                           | Def. 38.2.5.1(1, 13,<br>$\exists I(\wedge I(14, 15))$ )                                                                                                       |

|           |    |                          |                          |
|-----------|----|--------------------------|--------------------------|
| 1,2,3,8,9 | 17 | $u \star v \in A^\times$ | Def. 40.3.3.1(1, 13, 16) |
| 1,2,3,8   | 18 | $u \star v \in A^\times$ | $\exists E(7, 9, 17)$    |
| 1,2,3     | 19 | $u \star v \in A^\times$ | $\exists E(5, 8, 18)$    |

□

**Theorem 40.3.3.8 (Träger und Nichtleerheit der Einheitenmenge).**

**Mengen:**  $A, e, x, y, z$   
**Funktionen:**  $\star$

$$\text{Mon}(A, \star, e) \vdash e \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$$

|     |    |                                                                             |                                  |
|-----|----|-----------------------------------------------------------------------------|----------------------------------|
| 1   | 1  | $\text{Mon}(A, \star, e)$                                                   | $A$                              |
| 1   | 2  | $e \in A$                                                                   | Ax. 38.2.1.2(1)                  |
| 1   | 3  | $\text{Einh}_{A, \star, e}(e)$                                              | Th. 38.2.5.2(1)                  |
| 1   | 4  | $e \in A^\times$                                                            | Def. 40.3.3.1(1, 2, 3)           |
| 5   | 5  | $x \in A^\times$                                                            | $A$                              |
| 1,5 | 6  | $x \in A$                                                                   | Def. 40.3.3.1(1, 5)              |
| 1   | 7  | $A^\times \subseteq A$                                                      | Th. 3.5.1.1([5]:6)               |
| 1   | 8  | $A^\times \neq \emptyset$                                                   | Th. 3.6.3.5(4)                   |
| 1   | 9  | $A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$                       | Th. 3.16.3.1( $\wedge I(7, 8)$ ) |
| 1   | 10 | $e \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$ | $\wedge I(4, 9)$                 |

**Theorem 40.3.3.9 (Die Einheiten bilden eine Gruppe).**

**Mengen:**  $A, e, x, y, z$   
**Funktionen:**  $\star$

$$\text{Mon}(A, \star, e) \vdash \text{Grp}(A^\times, \star, e)$$

*Beweis.*

Halbgruppe der Einheiten:

|   |   |                                                                             |                                |
|---|---|-----------------------------------------------------------------------------|--------------------------------|
| 1 | 1 | $\text{Mon}(A, \star, e)$                                                   | $A$                            |
| 1 | 2 | $\text{HGr}(A, \star)$                                                      | Ax. 38.2.1.1(1)                |
| 1 | 3 | $e \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$ | Th. 40.3.3.8(1)                |
| 1 | 4 | $e \in A^\times$                                                            | $\wedge E1(3)$                 |
| 1 | 5 | $A^\times \subseteq A \wedge A^\times \neq \emptyset$                       | Th. 3.16.3.1( $\wedge E2(3)$ ) |

|       |    |                                                                          |                                                                                 |
|-------|----|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 6     | 6  | $x \in A^\times$                                                         | $A$                                                                             |
| 7     | 7  | $y \in A^\times$                                                         | $A$                                                                             |
| 1,6,7 | 8  | $x \star y \in A^\times$                                                 | Th. 40.3.3.7(1, 6, 7)                                                           |
| 1     | 9  | $\forall x \in A^\times \forall y \in A^\times (x \star y \in A^\times)$ | $\forall I(\rightarrow I(6, \forall I(\rightarrow I(7, 8))))$                   |
| 1     | 10 | $\text{UHGr}(A^\times; A, \star)$                                        | Th. 28.2.8.1(2, $\wedge I$<br>( $\wedge E1(5), \wedge I(\wedge E2$<br>(5), 9))) |
| 1     | 11 | $\text{HGr}(A^\times, \star)$                                            | Th. 28.2.8.2(2, 10)                                                             |

Neutrales Element und Inverse:

|      |    |                                                                                      |                                                  |
|------|----|--------------------------------------------------------------------------------------|--------------------------------------------------|
| 12   | 12 | $z \in A^\times$                                                                     | $A$                                              |
| 1,12 | 13 | $z \in A$                                                                            | Def. 40.3.3.1(1, 12)                             |
| 1,12 | 14 | $e \star z = z$                                                                      | Ax. 38.2.1.3(1, 13)                              |
| 1,12 | 15 | $z \star e = z$                                                                      | Ax. 38.2.1.4(1, 13)                              |
| 1    | 16 | $\forall z \in A^\times (e \star z = z \wedge z \star e = z)$                        | $\forall I(\rightarrow I(12, \wedge I(14, 15)))$ |
| 1    | 17 | $\text{Mon}(A^\times, \star, e)$                                                     | Def. 38.2.1.1(11, 4, 16)                         |
| 18   | 18 | $w \in A^\times$                                                                     | $A$                                              |
| 1,18 | 19 | $\exists y \in A^\times (w \star y = e \wedge y \star w = e)$                        | Th. 40.3.3.5(1, 18)                              |
| 1    | 20 | $\forall w \in A^\times \exists y \in A^\times (w \star y = e \wedge y \star w = e)$ | $\forall I(\rightarrow I(18, 19))$               |
| 1    | 21 | $\text{Grp}(A^\times, \star, e)$                                                     | Def. 40.2.1.1(17, 20)                            |

□

**Theorem 40.3.3.10 (Der volle Gruppenträger absorbiert nichtleere Teilmengen).**

**Mengen:**  $G, e, X, Y, x, y, z, a, b$

**Funktionen:**  $\star, \star_P$

$$\text{Grp}(G, \star, e), Y \in \mathcal{P}(G) \setminus \{\emptyset\} \vdash \\ Y \star_P G = G \wedge G \star_P Y = G$$

*Beweis.*

**Produktzeuge**

Th. 40.3.3.10(H1)

$$\text{HGr}(G, \star), X, Y \in \mathcal{P}(G) \setminus \{\emptyset\}, x \in X, y \in Y \\ \vdash x \star y \in X \star_P Y$$

|   |   |                        |     |
|---|---|------------------------|-----|
| 1 | 1 | $\text{HGr}(G, \star)$ | $A$ |
|---|---|------------------------|-----|

|           |    |                                                           |                                                   |
|-----------|----|-----------------------------------------------------------|---------------------------------------------------|
| 2         | 2  | $X \in \mathcal{P}(G) \setminus \{\emptyset\}$            | $A$                                               |
| 3         | 3  | $Y \in \mathcal{P}(G) \setminus \{\emptyset\}$            | $A$                                               |
| 4         | 4  | $x \in X$                                                 | $A$                                               |
| 5         | 5  | $y \in Y$                                                 | $A$                                               |
| 2         | 6  | $X \subseteq G$                                           | $\wedge E1(\text{Th. 3.16.3.1 (2)})$              |
| 3         | 7  | $Y \subseteq G$                                           | $\wedge E1(\text{Th. 3.16.3.1 (3)})$              |
| 2,4       | 8  | $x \in G$                                                 | $\text{Th. 3.5.2.6(6, 4)}$                        |
| 3,5       | 9  | $y \in G$                                                 | $\text{Th. 3.5.2.6(7, 5)}$                        |
| 1,2,3,4,5 | 10 | $x \star y \in G$                                         | $\text{Ax. 28.2.1.1(1, 8, 9)}$                    |
|           | 11 | $x \star y = x \star y$                                   | $= I$                                             |
| 5         | 12 | $\exists b \in Y (x \star y = x \star b)$                 | $\exists I(\wedge I(5, 11))$                      |
| 4,5       | 13 | $\exists a \in X \exists b \in Y (x \star y = a \star b)$ | $\exists I(\wedge I(4, 12))$                      |
| 1,2,3,4,5 | 14 | $x \star y \in X \star_{\mathcal{P}} Y$                   | $\text{Th. 28.2.4.1(1, 2, 3, } \wedge I(10, 13))$ |

Potenzträger und Zeuge der Nichtleerheit:

|        |    |                                                                           |                                                        |
|--------|----|---------------------------------------------------------------------------|--------------------------------------------------------|
| 1      | 1  | $\text{Grp}(G, \star, e)$                                                 | $A$                                                    |
| 2      | 2  | $Y \in \mathcal{P}(G) \setminus \{\emptyset\}$                            | $A$                                                    |
| 1      | 3  | $\text{HGr}(G, \star)$                                                    | $\text{Th. 40.2.2.1(1)}$                               |
| 1      | 4  | $\text{HGr}(\mathcal{P}(G) \setminus \{\emptyset\}, \star_{\mathcal{P}})$ | $\text{Th. 28.2.4.2(3)}$                               |
| 1      | 5  | $e \in G$                                                                 | $\text{Th. 40.2.2.4(1)}$                               |
| 1      | 6  | $G \neq \emptyset$                                                        | $\text{Th. 3.6.3.5(5)}$                                |
| 1      | 7  | $G \in \mathcal{P}(G) \setminus \{\emptyset\}$                            | $\text{Th. 3.16.3.1}(\wedge I(\text{Th. 3.5.3.1, 6}))$ |
| 2      | 8  | $Y \subseteq G \wedge Y \neq \emptyset$                                   | $\text{Th. 3.16.3.1(2)}$                               |
| 2      | 9  | $\exists y \in Y$                                                         | $\text{Th. 3.6.3.8}(\wedge E2(8))$                     |
| 10     | 10 | $y \in Y$                                                                 | $A$                                                    |
| 2,10   | 11 | $y \in G$                                                                 | $\text{Th. 3.5.2.6}(\wedge E1(8), 10)$                 |
| 1,2,10 | 12 | $y^{-1} \in G$                                                            | $\text{Th. 40.2.3.2(1, 11)}$                           |

Multiplikation von links:

|           |    |                                                                      |                                       |
|-----------|----|----------------------------------------------------------------------|---------------------------------------|
| 1,2       | 13 | $Y \star_{\mathcal{P}} G \in \mathcal{P}(G) \setminus \{\emptyset\}$ | $\text{Ax. 28.2.1.1(4, 2, 7)}$        |
| 1,2       | 14 | $Y \star_{\mathcal{P}} G \subseteq G$                                | $\wedge E1(\text{Th. 3.16.3.1 (13)})$ |
| 15        | 15 | $z \in G$                                                            | $A$                                   |
| 1,2,10,15 | 16 | $y^{-1} \star z \in G$                                               | $\text{Th. 40.2.2.2(1, 12, 15)}$      |
| 1,2,10    | 17 | $y \star y^{-1} = e$                                                 | $\text{Th. 40.2.3.3(1, 11)}$          |

|           |    |                                                       |                                                          |
|-----------|----|-------------------------------------------------------|----------------------------------------------------------|
| 1,2,10,15 | 18 | $(y \star y^{-1}) \star z = y \star (y^{-1} \star z)$ | Th. 40.2.2.3(1, 11, 12, 15)                              |
| 1,15      | 19 | $e \star z = z$                                       | Th. 40.2.2.5(1, 15)                                      |
| 1,2,10,15 | 20 | $y \star (y^{-1} \star z) = z$                        | $= E(17, 19, 18)$                                        |
| 1,2,10,15 | 21 | $z \in Y \star_{\mathcal{P}} G$                       | $= E(20,$<br>Th. 40.3.3.10( $H1$ )<br>(3, 2, 7, 10, 16)) |
| 1,2,10    | 22 | $G \subseteq Y \star_{\mathcal{P}} G$                 | Th. 3.5.1.1([15]:21)                                     |
| 1,2,10    | 23 | $Y \star_{\mathcal{P}} G = G$                         | Th. 3.5.3.2( $\wedge I(14,$<br>22))                      |

Multiplikation von rechts:

|           |    |                                                                      |                                                          |
|-----------|----|----------------------------------------------------------------------|----------------------------------------------------------|
| 1,2       | 24 | $G \star_{\mathcal{P}} Y \in \mathcal{P}(G) \setminus \{\emptyset\}$ | Ax. 28.2.1.1(4, 7, 2)                                    |
| 1,2       | 25 | $G \star_{\mathcal{P}} Y \subseteq G$                                | $\wedge E1$ (Th. 3.16.3.1<br>(24))                       |
| 26        | 26 | $z \in G$                                                            | $A$                                                      |
| 1,2,10,26 | 27 | $z \star y^{-1} \in G$                                               | Th. 40.2.2.2(1, 26, 12)                                  |
| 1,2,10    | 28 | $y^{-1} \star y = e$                                                 | Th. 40.2.3.4(1, 11)                                      |
| 1,2,10,26 | 29 | $(z \star y^{-1}) \star y = z \star (y^{-1} \star y)$                | Th. 40.2.2.3(1, 26, 12, 11)                              |
| 1,26      | 30 | $z \star e = z$                                                      | Th. 40.2.2.6(1, 26)                                      |
| 1,2,10,26 | 31 | $(z \star y^{-1}) \star y = z$                                       | $= E(28, 30, 29)$                                        |
| 1,2,10,26 | 32 | $z \in G \star_{\mathcal{P}} Y$                                      | $= E(31,$<br>Th. 40.3.3.10( $H1$ )<br>(3, 7, 2, 27, 10)) |
| 1,2,10    | 33 | $G \subseteq G \star_{\mathcal{P}} Y$                                | Th. 3.5.1.1([26]:32)                                     |
| 1,2,10    | 34 | $G \star_{\mathcal{P}} Y = G$                                        | Th. 3.5.3.2( $\wedge I(25,$<br>33))                      |
| 1,2,10    | 35 | $Y \star_{\mathcal{P}} G = G \wedge G \star_{\mathcal{P}} Y = G$     | $\wedge I(23, 34)$                                       |
| 1,2       | 36 | $Y \star_{\mathcal{P}} G = G \wedge G \star_{\mathcal{P}} Y = G$     | $\exists E(9, 10, 35)$                                   |

□

**Definition 40.3.3.2 (Einheitenstabiles Element).**

**Mengen:**  $M, e, z, u$

**Funktionen:**  $\star$

$$\text{UStab}_{M, \star, e}(z) := z \in M \wedge \forall u \in M \left( \text{Einh}_{M, \star, e}(u) \rightarrow (u \star z = z \wedge z \star u = z) \right)$$

**Theorem 40.3.3.11 (Isomorphismen transportieren Einheitenstabilität).**

**Mengen:**  $M, N, e, d, z, u, w$   
**Funktionen:**  $\star, \diamond, h$

$$\begin{aligned} & \text{Mon}(M, \star, e), \text{Mon}(N, \diamond, d), \\ & \text{HGrIso}(h; M, \star; N, \diamond), \text{UStab}_{M, \star, e}(z) \vdash \\ & \text{UStab}_{N, \diamond, d}(h(z)) \end{aligned}$$

*Beweis.*

Urbild einer Einheit:

|             |    |                                                                                   |                                       |
|-------------|----|-----------------------------------------------------------------------------------|---------------------------------------|
| 1           | 1  | $\text{Mon}(M, \star, e)$                                                         | $A$                                   |
| 2           | 2  | $\text{Mon}(N, \diamond, d)$                                                      | $A$                                   |
| 3           | 3  | $\text{HGrIso}(h; M, \star; N, \diamond)$                                         | $A$                                   |
| 4           | 4  | $\text{UStab}_{M, \star, e}(z)$                                                   | $A$                                   |
| 4           | 5  | $z \in M$                                                                         | Def. 40.3.3.2(4)                      |
| 3           | 6  | $h: M \xrightarrow{\sim} N$                                                       | Ax. 28.2.12.6(3)                      |
| 3           | 7  | $\text{HGrHom}(h; M, \star; N, \diamond)$                                         | Ax. 28.2.12.5(3)                      |
| 3,4         | 8  | $h(z) \in N$                                                                      | Th. 8.2.1.2(6, 5)                     |
| 9           | 9  | $w \in N$                                                                         | $A$                                   |
| 10          | 10 | $\text{Einh}_{N, \diamond, d}(w)$                                                 | $A$                                   |
| 3,9         | 11 | $\exists u \in M (h(u) = w)$                                                      | Def. 8.2.1.1(6, 9)                    |
| 12          | 12 | $u \in M \wedge h(u) = w$                                                         | $A$                                   |
| 12          | 13 | $u \in M$                                                                         | $\wedge E1(12)$                       |
| 12          | 14 | $h(u) = w$                                                                        | $\wedge E2(12)$                       |
| 1,2,3,12    | 15 | $\text{Einh}_{M, \star, e}(u) \leftrightarrow \text{Einh}_{N, \diamond, d}(h(u))$ | Th. 38.2.6.5(1, 2, 3, 13)             |
| 1,2,3,10,12 | 16 | $\text{Einh}_{M, \star, e}(u)$                                                    | $\leftrightarrow E2(15, = E(14, 10))$ |

Transport der beiden Gleichungen:

|               |    |                                      |                              |
|---------------|----|--------------------------------------|------------------------------|
| 1,2,3,4,10,12 | 17 | $u \star z = z \wedge z \star u = z$ | Def. 40.3.3.2(4, 13, 16)     |
| 3,4,12        | 18 | $h(u \star z) = h(u) \diamond h(z)$  | Ax. 28.2.12.3(7, 13, 5)      |
| 3,4,12        | 19 | $h(z \star u) = h(z) \diamond h(u)$  | Ax. 28.2.12.3(7, 5, 13)      |
| 1,2,3,4,10,12 | 20 | $w \diamond h(z) = h(z)$             | $= E(14, \wedge E1(17), 18)$ |
| 1,2,3,4,10,12 | 21 | $h(z) \diamond w = h(z)$             | $= E(14, \wedge E2(17), 19)$ |

|                   |    |                                                                                                                        |                                                      |
|-------------------|----|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 1,2,3,4,<br>10,12 | 22 | $w \diamond h(z) = h(z) \wedge h(z) \diamond w = h(z)$                                                                 | $\wedge I(20, 21)$                                   |
| 1,2,3,4,9,<br>10  | 23 | $w \diamond h(z) = h(z) \wedge h(z) \diamond w = h(z)$                                                                 | $\exists E(11, 12, 22)$                              |
| 1,2,3,4           | 24 | $\forall w \in N (\text{Einh}_{N, \diamond, d}(w) \rightarrow (w \diamond h(z) = h(z) \wedge h(z) \diamond w = h(z)))$ | $\forall I(\rightarrow I(9, \rightarrow I(10, 23)))$ |
| 1,2,3,4           | 25 | $\text{UStab}_{N, \diamond, d}(h(z))$                                                                                  | Def. 40.3.3.2(8, 24)                                 |

□

**Theorem 40.3.3.12 (Die Einheitenmenge ist im Potenzmonoid einheitenstabil).**

**Mengen:**  $A, e, u, W$   
**Funktionen:**  $\star, \star_{\mathcal{P}}$

$$\text{Mon}(A, \star, e) \vdash \text{UStab}_{\mathcal{P}(A) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\}}(A^\times)$$

|       |    |                                                                                                                                                                                                                                                       |                                                            |
|-------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1     | 1  | $\text{Mon}(A, \star, e)$                                                                                                                                                                                                                             | $A$                                                        |
| 1     | 2  | $e \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$                                                                                                                                                                           | Th. 40.3.3.8(1)                                            |
| 1     | 3  | $A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$                                                                                                                                                                                                 | $\wedge E2(2)$                                             |
| 1     | 4  | $\text{Grp}(A^\times, \star, e)$                                                                                                                                                                                                                      | Th. 40.3.3.9(1)                                            |
| 5     | 5  | $W \in \mathcal{P}(A) \setminus \{\emptyset\}$                                                                                                                                                                                                        | $A$                                                        |
| 6     | 6  | $\text{Einh}_{\mathcal{P}(A) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\}}(W)$                                                                                                                                                                 | $A$                                                        |
| 1,5,6 | 7  | $\exists u \in A (W = \{u\} \wedge \text{Einh}_{A, \star, e}(u))$                                                                                                                                                                                     | $\leftrightarrow E1(\text{Th. 38.2.5.1}(1, 5), 6)$         |
| 8     | 8  | $u \in A \wedge (W = \{u\} \wedge \text{Einh}_{A, \star, e}(u))$                                                                                                                                                                                      | $A$                                                        |
| 1,8   | 9  | $u \in A^\times$                                                                                                                                                                                                                                      | Def. 40.3.3.1(1, $\wedge E1(8), \wedge E2(\wedge E2(8))$ ) |
| 1,8   | 10 | $\{u\} \in \mathcal{P}(A^\times) \setminus \{\emptyset\}$                                                                                                                                                                                             | Th. 38.2.3.1(9)                                            |
| 1,8   | 11 | $\{u\} \star_{\mathcal{P}} A^\times = A^\times \wedge A^\times \star_{\mathcal{P}} \{u\} = A^\times$                                                                                                                                                  | Th. 40.3.3.10(4, 10)                                       |
| 1,8   | 12 | $W \star_{\mathcal{P}} A^\times = A^\times \wedge A^\times \star_{\mathcal{P}} W = A^\times$                                                                                                                                                          | $= E(\wedge E1(\wedge E2(8)), 11)$                         |
| 1,5,6 | 13 | $W \star_{\mathcal{P}} A^\times = A^\times \wedge A^\times \star_{\mathcal{P}} W = A^\times$                                                                                                                                                          | $\exists E(7, 8, 12)$                                      |
| 1     | 14 | $\forall W \in \mathcal{P}(A) \setminus \{\emptyset\} (\text{Einh}_{\mathcal{P}(A) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\}}(W) \rightarrow (W \star_{\mathcal{P}} A^\times = A^\times \wedge A^\times \star_{\mathcal{P}} W = A^\times))$ | $\forall I(\rightarrow I(5, \rightarrow I(6, 13)))$        |
| 1     | 15 | $\text{UStab}_{\mathcal{P}(A) \setminus \{\emptyset\}, \star_{\mathcal{P}}, \{e\}}(A^\times)$                                                                                                                                                         | Def. 40.3.3.2(3, 14)                                       |

**Theorem 40.3.3.13 (Einheitenstabilität liefert Absorption durch die Einheitenmenge).**

**Mengen:**  $A, e, X, x, z, u$   
**Funktionen:**  $\star, \star_{\mathcal{P}}$   
**Abkürzungen:**  $U = A^\times, P = \mathcal{P}(A) \setminus \{\emptyset\}$

$$\begin{aligned} & \text{Mon}(A, \star, e), \quad X \in \mathcal{P}(A) \setminus \{\emptyset\}, \\ & \text{UStab}_{\mathcal{P}(A) \setminus \{\emptyset\}, \star_P, \{e\}}(X) \vdash \\ & A^\times \star_P X = X \wedge X \star_P A^\times = X \end{aligned}$$

*Beweis.*

Träger:

|   |   |                                         |                 |
|---|---|-----------------------------------------|-----------------|
| 1 | 1 | $\text{Mon}(A, \star, e)$               | $A$             |
| 2 | 2 | $X \in P$                               | $A$             |
| 3 | 3 | $\text{UStab}_{P, \star_P, \{e\}}(X)$   | $A$             |
| 1 | 4 | $\text{HGr}(A, \star)$                  | Ax. 38.2.1.1(1) |
| 1 | 5 | $e \in U \wedge U \in P$                | Th. 40.3.3.8(1) |
| 1 | 6 | $e \in U$                               | $\wedge E1(5)$  |
| 1 | 7 | $U \in P$                               | $\wedge E2(5)$  |
| 2 | 8 | $X \subseteq A \wedge X \neq \emptyset$ | Th. 3.16.3.1(2) |

Linksprodukt mit der Einheitenmenge:

|            |    |                                                                  |                                                                                                   |
|------------|----|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 9          | 9  | $z \in U \star_P X$                                              | $A$                                                                                               |
| 1,2,9      | 10 | $z \in A \wedge \exists u \in U \exists x \in X (z = u \star x)$ | Th. 28.2.4.1(4, 7, 2, 9)                                                                          |
| 1,2,9      | 11 | $z \in A$                                                        | $\wedge E1(10)$                                                                                   |
| 1,2,9      | 12 | $\exists u \in U \exists x \in X (z = u \star x)$                | $\wedge E2(10)$                                                                                   |
| 13         | 13 | $u \in U \wedge (x \in X \wedge z = u \star x)$                  | $A$                                                                                               |
| 1,13       | 14 | $u \in A$                                                        | Def. 40.3.3.1(1, $\wedge E1(13)$ )                                                                |
| 1,13       | 15 | $\text{Einh}_{A, \star, e}(u)$                                   | Def. 40.3.3.1(1, $\wedge E1(13)$ )                                                                |
| 1,13       | 16 | $\{u\} \in P$                                                    | Th. 38.2.3.1(14)                                                                                  |
| 1,13       | 17 | $\text{Einh}_{P, \star_P, \{e\}}(\{u\})$                         | $\leftrightarrow E2(\text{Th. 38.2.5.1}(1, 16), \exists I(\wedge I(14, \wedge I(= I, 15))))$      |
| 1,3,13     | 18 | $\{u\} \star_P X = X$                                            | Def. 40.3.3.2(3, 16, 17)                                                                          |
|            | 19 | $u \in \{u\}$                                                    | Th. 3.11.3.8                                                                                      |
| 1,2,9,13   | 20 | $z \in \{u\} \star_P X$                                          | $= E(\wedge E2(\wedge E2(13)), \text{Th. 40.3.3.10}(H1)(4, 16, 2, 19, \wedge E1(\wedge E2(13))))$ |
| 1,2,3,9,13 | 21 | $z \in X$                                                        | $= E(18, 20)$                                                                                     |
| 1,2,3,9    | 22 | $z \in X$                                                        | $\exists E^*(12, 13, 21)$                                                                         |

|        |    |                           |                                                         |
|--------|----|---------------------------|---------------------------------------------------------|
| 1,2,3  | 23 | $U \star_P X \subseteq X$ | Th. 3.5.1.1([9]:22)                                     |
| 24     | 24 | $x \in X$                 | $A$                                                     |
| 2,24   | 25 | $x \in A$                 | Th. 3.5.2.6( $\wedge E1(8)$ ,<br>24)                    |
| 1,2,24 | 26 | $e \star x = x$           | Ax. 38.2.1.3(1, 25)                                     |
| 1,2,24 | 27 | $x \in U \star_P X$       | $= E(26,$<br>Th. 40.3.3.10( $H1$ )<br>(4, 7, 2, 6, 24)) |
| 1,2    | 28 | $X \subseteq U \star_P X$ | Th. 3.5.1.1([24]:27)                                    |
| 1,2,3  | 29 | $U \star_P X = X$         | Th. 3.5.3.2( $\wedge I(23,$<br>28))                     |

Rechtsprodukt mit der Einheitenmenge:

|                 |    |                                                                  |                                                                                                           |
|-----------------|----|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 30              | 30 | $z \in X \star_P U$                                              | $A$                                                                                                       |
| 1,2,30          | 31 | $z \in A \wedge \exists x \in X \exists u \in U (z = x \star u)$ | Th. 28.2.4.1(4, 2, 7,<br>30)                                                                              |
| 1,2,30          | 32 | $z \in A$                                                        | $\wedge E1(31)$                                                                                           |
| 1,2,30          | 33 | $\exists x \in X \exists u \in U (z = x \star u)$                | $\wedge E2(31)$                                                                                           |
| 34              | 34 | $x \in X \wedge (u \in U \wedge z = x \star u)$                  | $A$                                                                                                       |
| 1,34            | 35 | $u \in A$                                                        | Def. 40.3.3.1(1, $\wedge E1$<br>( $\wedge E2(34)$ ))                                                      |
| 1,34            | 36 | $\text{Einh}_{A, \star, e}(u)$                                   | Def. 40.3.3.1(1, $\wedge E1$<br>( $\wedge E2(34)$ ))                                                      |
| 1,34            | 37 | $\{u\} \in P$                                                    | Th. 38.2.3.1(35)                                                                                          |
| 1,34            | 38 | $\text{Einh}_{P, \star_P, \{e\}}(\{u\})$                         | $\leftrightarrow E2(\text{Th. 38.2.5.1}$<br>(1, 37), $\exists I(\wedge I(35, \wedge I$<br>( $= I, 36))))$ |
| 1,3,34          | 39 | $X \star_P \{u\} = X$                                            | Def. 40.3.3.2(3, 37,<br>38)                                                                               |
|                 | 40 | $u \in \{u\}$                                                    | Th. 3.11.3.8                                                                                              |
| 1,2,30,34       | 41 | $z \in X \star_P \{u\}$                                          | $= E(\wedge E2(\wedge E2$<br>(34)),<br>Th. 40.3.3.10( $H1$ )<br>(4, 2, 37, $\wedge E1(34)$ ,<br>40))      |
| 1,2,3,30,<br>34 | 42 | $z \in X$                                                        | $= E(39, 41)$                                                                                             |
| 1,2,3,30        | 43 | $z \in X$                                                        | $\exists E^*(33, 34, 42)$                                                                                 |
| 1,2,3           | 44 | $X \star_P U \subseteq X$                                        | Th. 3.5.1.1([30]:43)                                                                                      |
| 45              | 45 | $x \in X$                                                        | $A$                                                                                                       |
| 2,45            | 46 | $x \in A$                                                        | Th. 3.5.2.6( $\wedge E1(8)$ ,<br>45)                                                                      |
| 1,2,45          | 47 | $x \star e = x$                                                  | Ax. 38.2.1.4(1, 46)                                                                                       |

|        |    |                                                                  |                                                         |
|--------|----|------------------------------------------------------------------|---------------------------------------------------------|
| 1,2,45 | 48 | $x \in X \star_{\mathcal{P}} U$                                  | $= E(47,$<br>Th. 40.3.3.10( $H1$ )<br>(4, 2, 7, 45, 6)) |
| 1,2    | 49 | $X \subseteq X \star_{\mathcal{P}} U$                            | Th. 3.5.1.1([45]:48)                                    |
| 1,2,3  | 50 | $X \star_{\mathcal{P}} U = X$                                    | Th. 3.5.3.2( $\wedge I(44,$<br>49))                     |
| 1,2,3  | 51 | $U \star_{\mathcal{P}} X = X \wedge X \star_{\mathcal{P}} U = X$ | $\wedge I(29, 50)$                                      |

□

**Theorem 40.3.3.14 (Produktkriterium für Teilmengen der Einheitsmenge).**

**Mengen:**  $A, e, Y, y$

**Funktionen:**  $\star, \star_{\mathcal{P}}$

$$\begin{aligned} & \text{Mon}(A, \star, e), Y \in \mathcal{P}(A) \setminus \{\emptyset\} \vdash \\ & Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\} \leftrightarrow Y \star_{\mathcal{P}} A^\times = A^\times \end{aligned}$$

*Beweis.*

Nichtleere Teilmengen der Einheitsmenge:

|     |   |                                                                                                             |                                  |
|-----|---|-------------------------------------------------------------------------------------------------------------|----------------------------------|
| 1   | 1 | $\text{Mon}(A, \star, e)$                                                                                   | $A$                              |
| 2   | 2 | $Y \in \mathcal{P}(A) \setminus \{\emptyset\}$                                                              | $A$                              |
| 1   | 3 | $e \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$                                 | Th. 40.3.3.8(1)                  |
| 1   | 4 | $\text{Grp}(A^\times, \star, e)$                                                                            | Th. 40.3.3.9(1)                  |
| 5   | 5 | $Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\}$                                                       | $A$                              |
| 1,5 | 6 | $Y \star_{\mathcal{P}} A^\times = A^\times \wedge A^\times \star_{\mathcal{P}} Y = A^\times$                | Th. 40.3.3.10(4, 5)              |
| 1   | 7 | $Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\} \rightarrow Y \star_{\mathcal{P}} A^\times = A^\times$ | $\rightarrow I(5, \wedge E1(6))$ |

Rückschluss mit dem neutralen Element:

|         |    |                                             |                                                                                       |
|---------|----|---------------------------------------------|---------------------------------------------------------------------------------------|
| 8       | 8  | $Y \star_{\mathcal{P}} A^\times = A^\times$ | $A$                                                                                   |
| 9       | 9  | $y \in Y$                                   | $A$                                                                                   |
| 2       | 10 | $Y \subseteq A \wedge Y \neq \emptyset$     | Th. 3.16.3.1(2)                                                                       |
| 2,9     | 11 | $y \in A$                                   | Th. 3.5.2.6( $\wedge E1$<br>(10), 9)                                                  |
| 1       | 12 | $\text{HGr}(A, \star)$                      | Ax. 38.2.1.1(1)                                                                       |
| 1,2,9   | 13 | $y \star e = y$                             | Ax. 38.2.1.4(1, 11)                                                                   |
| 1,2,9   | 14 | $y \in Y \star_{\mathcal{P}} A^\times$      | $= E(13,$<br>Th. 40.3.3.10( $H1$ )<br>(12, 2, $\wedge E2(3)$ , 9,<br>$\wedge E1(3)))$ |
| 1,2,8,9 | 15 | $y \in A^\times$                            | $= E(8, 14)$                                                                          |

|       |    |                                                                                                                 |                                               |
|-------|----|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1,2,8 | 16 | $Y \subseteq A^\times$                                                                                          | Th. 3.5.1.1([9]:15)                           |
| 1,2,8 | 17 | $Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\}$                                                           | Th. 3.16.3.1( $\wedge I(16, \wedge E2(10))$ ) |
| 1,2   | 18 | $Y \star_{\mathcal{P}} A^\times = A^\times \rightarrow Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\}$     | $\rightarrow I(8, 17)$                        |
| 1,2   | 19 | $Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\} \leftrightarrow Y \star_{\mathcal{P}} A^\times = A^\times$ | $\leftrightarrow I(7, 18)$                    |

□

**Theorem 40.3.3.15 (Bild der gesamten Einheitenmenge).**

**Mengen:**  $A, B, e_A, e_B$   
**Funktionen:**  $\star, \diamond, \star_{\mathcal{P}}, \diamond_{\mathcal{P}}, F$   
**Abkürzungen:**  $U = A^\times, V = B^\times$   
 $P = \mathcal{P}(A) \setminus \{\emptyset\}, Q = \mathcal{P}(B) \setminus \{\emptyset\}$

$$\text{Mon}(A, \star, e_A), \text{Mon}(B, \diamond, e_B),$$

$$\text{HGrIso}(F; \mathcal{P}(A) \setminus \{\emptyset\}, \star_{\mathcal{P}}; \mathcal{P}(B) \setminus \{\emptyset\}, \diamond_{\mathcal{P}}) \vdash F(A^\times) = B^\times$$

*Beweis.*

Einheitenstabilität der beiden Bilder:

|       |    |                                                                            |                             |
|-------|----|----------------------------------------------------------------------------|-----------------------------|
| 1     | 1  | $\text{Mon}(A, \star, e_A)$                                                | $A$                         |
| 2     | 2  | $\text{Mon}(B, \diamond, e_B)$                                             | $A$                         |
| 3     | 3  | $\text{HGrIso}(F; P, \star_{\mathcal{P}}; Q, \diamond_{\mathcal{P}})$      | $A$                         |
| 1     | 4  | $\text{Mon}(P, \star_{\mathcal{P}}, \{e_A\})$                              | Th. 38.2.2.3(1)             |
| 2     | 5  | $\text{Mon}(Q, \diamond_{\mathcal{P}}, \{e_B\})$                           | Th. 38.2.2.3(2)             |
| 1     | 6  | $e_A \in U \wedge U \in P$                                                 | Th. 40.3.3.8(1)             |
| 2     | 7  | $e_B \in V \wedge V \in Q$                                                 | Th. 40.3.3.8(2)             |
| 1     | 8  | $U \in P$                                                                  | $\wedge E2(6)$              |
| 2     | 9  | $V \in Q$                                                                  | $\wedge E2(7)$              |
| 1     | 10 | $\text{UStab}_{P, \star_{\mathcal{P}}, \{e_A\}}(U)$                        | Th. 40.3.3.12(1)            |
| 2     | 11 | $\text{UStab}_{Q, \diamond_{\mathcal{P}}, \{e_B\}}(V)$                     | Th. 40.3.3.12(2)            |
| 3     | 12 | $\text{HGrIso}(F^{-1}; Q, \diamond_{\mathcal{P}}; P, \star_{\mathcal{P}})$ | Th. 28.2.15.1(3)            |
| 1,2,3 | 13 | $\text{UStab}_{Q, \diamond_{\mathcal{P}}, \{e_B\}}(F(U))$                  | Th. 40.3.3.11(4, 5, 3, 10)  |
| 1,2,3 | 14 | $\text{UStab}_{P, \star_{\mathcal{P}}, \{e_A\}}(F^{-1}(V))$                | Th. 40.3.3.11(5, 4, 12, 11) |

Gleichheit der Einheitenmengenbilder:

|     |    |                                  |                    |
|-----|----|----------------------------------|--------------------|
| 3   | 15 | $F: P \xrightarrow{\sim} Q$      | Ax. 28.2.12.6(3)   |
| 3   | 16 | $F^{-1}: Q \xrightarrow{\sim} P$ | Ax. 28.2.12.6(12)  |
| 1,3 | 17 | $F(U) \in Q$                     | Th. 8.2.1.2(15, 8) |

|       |    |                                                                          |                                       |
|-------|----|--------------------------------------------------------------------------|---------------------------------------|
| 2,3   | 18 | $F^{-1}(V) \in P$                                                        | Th. 8.2.1.2(16, 9)                    |
| 1,2,3 | 19 | $V \diamond_P F(U) = F(U) \wedge F(U) \diamond_P V = F(U)$               | Th. 40.3.3.13(2, 17, 13)              |
| 1,2,3 | 20 | $U \star_P F^{-1}(V) = F^{-1}(V) \wedge F^{-1}(V) \star_P U = F^{-1}(V)$ | Th. 40.3.3.13(1, 18, 14)              |
| 2,3   | 21 | $F(F^{-1}(V)) = V$                                                       | Th. 8.3.3.6(15, 9)                    |
| 3     | 22 | $\text{HGrHom}(F; P, \star_P; Q, \diamond_P)$                            | Ax. 28.2.12.5(3)                      |
| 1,2,3 | 23 | $F(U \star_P F^{-1}(V)) = F(U) \diamond_P F(F^{-1}(V))$                  | Ax. 28.2.12.3(22, 8, 18)              |
| 1,2,3 | 24 | $V = F(U) \diamond_P V$                                                  | $= E(\wedge E1(20), 21, 23)$          |
| 1,2,3 | 25 | $F(U) = V$                                                               | Th. 2.9.1.1(Tr.(24, $\wedge E2(19)))$ |

□

**Theorem 40.3.3.16 (Bilder nichtleerer Teilmengen der Einheitsmenge).**

**Mengen:**  $A, B, e_A, e_B, Y$

**Funktionen:**  $\star, \diamond, \star_P, \diamond_P, F$

$\text{Mon}(A, \star, e_A), \text{Mon}(B, \diamond, e_B), \text{HGrIso}(F; \mathcal{P}(A) \setminus \{\emptyset\}, \star_P; \mathcal{P}(B) \setminus \{\emptyset\}, \diamond_P),$   
 $F(A^\times) = B^\times, Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\} \vdash F(Y) \in \mathcal{P}(B^\times) \setminus \{\emptyset\}$

|         |    |                                                                                                                         |                                                     |
|---------|----|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1       | 1  | $\text{Mon}(A, \star, e_A)$                                                                                             | $A$                                                 |
| 2       | 2  | $\text{Mon}(B, \diamond, e_B)$                                                                                          | $A$                                                 |
| 3       | 3  | $\text{HGrIso}(F; \mathcal{P}(A) \setminus \{\emptyset\}, \star_P; \mathcal{P}(B) \setminus \{\emptyset\}, \diamond_P)$ | $A$                                                 |
| 4       | 4  | $F(A^\times) = B^\times$                                                                                                | $A$                                                 |
| 5       | 5  | $Y \in \mathcal{P}(A^\times) \setminus \{\emptyset\}$                                                                   | $A$                                                 |
| 1       | 6  | $e_A \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$                                           | Th. 40.3.3.8(1)                                     |
| 1       | 7  | $A^\times \subseteq A$                                                                                                  | $\wedge E1(\text{Th. 3.16.3.1}(\wedge E2(6)))$      |
| 1,5     | 8  | $Y \in \mathcal{P}(A) \setminus \{\emptyset\}$                                                                          | Th. 3.16.3.2(7, 5)                                  |
| 3       | 9  | $F: \mathcal{P}(A) \setminus \{\emptyset\} \xrightarrow{\sim} \mathcal{P}(B) \setminus \{\emptyset\}$                   | Ax. 28.2.12.6(3)                                    |
| 1,3,5   | 10 | $F(Y) \in \mathcal{P}(B) \setminus \{\emptyset\}$                                                                       | Th. 8.2.1.2(9, 8)                                   |
| 1,5     | 11 | $Y \star_P A^\times = A^\times$                                                                                         | $\leftrightarrow E1(\text{Th. 40.3.3.14}(1, 8), 5)$ |
| 3       | 12 | $\text{HGrHom}(F; \mathcal{P}(A) \setminus \{\emptyset\}, \star_P; \mathcal{P}(B) \setminus \{\emptyset\}, \diamond_P)$ | Ax. 28.2.12.5(3)                                    |
| 1,3,5   | 13 | $F(Y \star_P A^\times) = F(Y) \diamond_P F(A^\times)$                                                                   | Ax. 28.2.12.3(12, 8, $\wedge E2(6)$ )               |
| 1,3,4,5 | 14 | $F(Y) \diamond_P B^\times = B^\times$                                                                                   | $= E(11, 4, 13)$                                    |

1,2,3,4,5 15  $F(Y) \in \mathcal{P}(B^\times) \setminus \{\emptyset\}$

$\leftrightarrow E2(\text{Th. 40.3.3.14}$   
(2, 10), 14)

**Theorem 40.3.3.17 (Alle Gruppenelemente sind Einheiten).**

**Mengen:**  $A, e, x, y, z$

**Funktionen:**  $\star$

$$\text{Grp}(A, \star, e) \vdash A^\times = A$$

|     |   |                                                                             |                                                        |
|-----|---|-----------------------------------------------------------------------------|--------------------------------------------------------|
| 1   | 1 | $\text{Grp}(A, \star, e)$                                                   | $A$                                                    |
| 1   | 2 | $\text{Mon}(A, \star, e)$                                                   | Ax. 40.2.1.2(1)                                        |
| 1   | 3 | $e \in A^\times \wedge A^\times \in \mathcal{P}(A) \setminus \{\emptyset\}$ | Th. 40.3.3.8(2)                                        |
| 1   | 4 | $A^\times \subseteq A$                                                      | $\wedge E1(\text{Th. 3.16.3.1}$<br>( $\wedge E2(3)$ )) |
| 5   | 5 | $x \in A$                                                                   | $A$                                                    |
| 1,5 | 6 | $\text{Einh}_{A, \star, e}(x)$                                              | Th. 40.3.3.1(1, 5)                                     |
| 1,5 | 7 | $x \in A^\times$                                                            | Def. 40.3.3.1(2, 5, 6)                                 |
| 1   | 8 | $A \subseteq A^\times$                                                      | Th. 3.5.1.1([5]:7)                                     |
| 1   | 9 | $A^\times = A$                                                              | Th. 3.5.3.2( $\wedge I(4, 8)$ )                        |

**Theorem 40.3.3.18 (Große Potenzhalbgruppen rekonstruieren die Einheitengruppen).**

**Mengen:**  $A, B, A^\times, B^\times, X, Y, e_A, e_B$   
**Funktionen:**  $\star, \diamond, \star_P, \diamond_P, F$   
**Abkürzungen:**  $U = A^\times, V = B^\times$   
 $P = \mathcal{P}(A) \setminus \{\emptyset\}, Q = \mathcal{P}(B) \setminus \{\emptyset\}$   
 $R = F \upharpoonright_{\mathcal{P}(U) \setminus \{\emptyset\}}$

$$\begin{aligned} & \text{Mon}(A, \star, e_A), \text{Mon}(B, \diamond, e_B), \\ & \text{HGrIso}(F; \mathcal{P}(A) \setminus \{\emptyset\}, \star_P; \mathcal{P}(B) \setminus \{\emptyset\}, \diamond_P) \vdash \\ & F(A^\times) = B^\times \wedge F[\mathcal{P}(A^\times) \setminus \{\emptyset\}] = \mathcal{P}(B^\times) \setminus \{\emptyset\} \end{aligned}$$

*Beweis.*

Bild der Einheitenmenge und Umkehrabbildung:

|       |   |                                                    |                        |
|-------|---|----------------------------------------------------|------------------------|
| 1     | 1 | $\text{Mon}(A, \star, e_A)$                        | $A$                    |
| 2     | 2 | $\text{Mon}(B, \diamond, e_B)$                     | $A$                    |
| 3     | 3 | $\text{HGrIso}(F; P, \star_P; Q, \diamond_P)$      | $A$                    |
| 1,2,3 | 4 | $F(U) = V$                                         | Th. 40.3.3.15(1, 2, 3) |
| 3     | 5 | $\text{HGrIso}(F^{-1}; Q, \diamond_P; P, \star_P)$ | Th. 28.2.15.1(3)       |
| 1,2,3 | 6 | $F^{-1}(V) = U$                                    | Th. 40.3.3.15(2, 1, 5) |
| 3     | 7 | $F: P \xrightarrow{\sim} Q$                        | Ax. 28.2.12.6(3)       |

Punktweiser Transport in beiden Richtungen:

|            |    |                                                                                                                                                                  |                               |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 8          | 8  | $Y \in P$                                                                                                                                                        | $A$                           |
| 3,8        | 9  | $F(Y) \in Q$                                                                                                                                                     | Th. 8.2.1.2(7, 8)             |
| 10         | 10 | $Y \in \mathcal{P}(U) \setminus \{\emptyset\}$                                                                                                                   | $A$                           |
| 1,2,3,10   | 11 | $F(Y) \in \mathcal{P}(V) \setminus \{\emptyset\}$                                                                                                                | Th. 40.3.3.16(1, 2, 3, 4, 10) |
| 1,2,3      | 12 | $Y \in \mathcal{P}(U) \setminus \{\emptyset\} \rightarrow F(Y) \in \mathcal{P}(V) \setminus \{\emptyset\}$                                                       | $\rightarrow I(10, 11)$       |
| 13         | 13 | $F(Y) \in \mathcal{P}(V) \setminus \{\emptyset\}$                                                                                                                | $A$                           |
| 1,2,3,13   | 14 | $F^{-1}(F(Y)) \in \mathcal{P}(U) \setminus \{\emptyset\}$                                                                                                        | Th. 40.3.3.16(2, 1, 5, 6, 13) |
| 3,8        | 15 | $F^{-1}(F(Y)) = Y$                                                                                                                                               | Th. 8.3.3.5(7, 8)             |
| 1,2,3,8,13 | 16 | $Y \in \mathcal{P}(U) \setminus \{\emptyset\}$                                                                                                                   | $= E(15, 14)$                 |
| 1,2,3,8    | 17 | $F(Y) \in \mathcal{P}(V) \setminus \{\emptyset\} \rightarrow Y \in \mathcal{P}(U) \setminus \{\emptyset\}$                                                       | $\rightarrow I(13, 16)$       |
| 1,2,3,8    | 18 | $Y \in \mathcal{P}(U) \setminus \{\emptyset\} \leftrightarrow F(Y) \in \mathcal{P}(V) \setminus \{\emptyset\}$                                                   | $\leftrightarrow I(12, 17)$   |
| 1,2,3      | 19 | $\forall Y \in P (Y \in \mathcal{P}(U) \setminus \{\emptyset\} \leftrightarrow F(Y) \in \mathcal{P}(V) \setminus \{\emptyset\}) \forall I(\rightarrow I(8, 18))$ |                               |

Bild der ganzen Teilmengenfamilie:

|       |    |                                                                                                       |                                                 |
|-------|----|-------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1     | 20 | $e_A \in U \wedge U \in P$                                                                            | Th. 40.3.3.8(1)                                 |
| 1     | 21 | $U \subseteq A$                                                                                       | $\wedge E1(\text{Th. 3.16.3.1}(\wedge E2(20)))$ |
| 22    | 22 | $Z \in \mathcal{P}(U) \setminus \{\emptyset\}$                                                        | $A$                                             |
| 1,22  | 23 | $Z \in P$                                                                                             | Th. 3.16.3.2(21, 22)                            |
| 1     | 24 | $\mathcal{P}(U) \setminus \{\emptyset\} \subseteq P$                                                  | Th. 3.5.1.1([22]:23)                            |
| 2     | 25 | $e_B \in V \wedge V \in Q$                                                                            | Th. 40.3.3.8(2)                                 |
| 2     | 26 | $V \subseteq B$                                                                                       | $\wedge E1(\text{Th. 3.16.3.1}(\wedge E2(25)))$ |
| 27    | 27 | $Z \in \mathcal{P}(V) \setminus \{\emptyset\}$                                                        | $A$                                             |
| 2,27  | 28 | $Z \in Q$                                                                                             | Th. 3.16.3.2(26, 27)                            |
| 2     | 29 | $\mathcal{P}(V) \setminus \{\emptyset\} \subseteq Q$                                                  | Th. 3.5.1.1([27]:28)                            |
| 1,2,3 | 30 | $R: \mathcal{P}(U) \setminus \{\emptyset\} \xrightarrow{\sim} \mathcal{P}(V) \setminus \{\emptyset\}$ | Th. 8.3.2.2(H4)(7, 24, 29, 19)                  |
| 1,2,3 | 31 | $R[\mathcal{P}(U) \setminus \{\emptyset\}] = \mathcal{P}(V) \setminus \{\emptyset\}$                  | Th. 8.2.2.1(30)                                 |
| 3     | 32 | $F: P \rightarrow Q$                                                                                  | Def. 8.2.1.1(7)                                 |
| 1,3   | 33 | $R[\mathcal{P}(U) \setminus \{\emptyset\}] = F[\mathcal{P}(U) \setminus \{\emptyset\}]$               | Th. 5.3.2.16(24, 32)                            |
| 1,2,3 | 34 | $F[\mathcal{P}(U) \setminus \{\emptyset\}] = \mathcal{P}(V) \setminus \{\emptyset\}$                  | Th. 2.9.1.4(33, 31)                             |
| 1,2,3 | 35 | $F(U) = V \wedge F[\mathcal{P}(U) \setminus \{\emptyset\}] = \mathcal{P}(V) \setminus \{\emptyset\}$  | $\wedge I(4, 34)$                               |

□

Der folgende Gruppenstarrheitssatz geht in dieser Form auf Liu–Tringali (2026), Satz 1.3 zurück. Im Unterschied zum zuvor benötigten endlichen Argument kommt er ohne jede Kardinalitätsannahme aus.

**Theorem 40.3.3.19 (Einseitige Gruppenstarrheit der großen Potenzhalbgruppe).**

**Mengen:**  $A, B, A^\times, B^\times, E, X, Y, e_A, e_B$   
**Funktionen:**  $\star, \diamond, \star_{\mathcal{P}}, \diamond_{\mathcal{P}}, F, f$   
**Abkürzungen:**  $U = A^\times, V = B^\times$   
 $P = \mathcal{P}(A) \setminus \{\emptyset\}, Q = \mathcal{P}(B) \setminus \{\emptyset\}$

$\text{Grp}(A, \star, e_A), \text{HGr}(B, \diamond),$   
 $\text{HGrIso}(F; \mathcal{P}(A) \setminus \{\emptyset\}, \star_{\mathcal{P}}; \mathcal{P}(B) \setminus \{\emptyset\}, \diamond_{\mathcal{P}}) \vdash$   
 $\exists e_B \in B \exists f (\text{Grp}(B, \diamond, e_B) \wedge \text{GrpIso}(f; A, \star, e_A; B, \diamond, e_B))$

*Beweis.*

Monoidstruktur des Zielträgers:

|   |   |                                                                       |     |
|---|---|-----------------------------------------------------------------------|-----|
| 1 | 1 | $\text{Grp}(A, \star, e_A)$                                           | $A$ |
| 2 | 2 | $\text{HGr}(B, \diamond)$                                             | $A$ |
| 3 | 3 | $\text{HGrIso}(F; P, \star_{\mathcal{P}}; Q, \diamond_{\mathcal{P}})$ | $A$ |

|       |    |                                                                                |                           |
|-------|----|--------------------------------------------------------------------------------|---------------------------|
| 1     | 4  | $\text{Mon}(A, \star, e_A)$                                                    | Ax. 40.2.1.2(1)           |
| 1     | 5  | $\text{Mon}(P, \star_P, \{e_A\})$                                              | Th. 38.2.2.3(4)           |
| 1,3   | 6  | $\text{Mon}(Q, \diamond_P, F(\{e_A\}))$                                        | Th. 38.2.6.2(5, 3)        |
| 1,2,3 | 7  | $\exists e_B \in B (F(\{e_A\}) = \{e_B\} \wedge \text{Mon}(B, \diamond, e_B))$ | Th. 38.2.2.4(2, 6)        |
| 8     | 8  | $e_B \in B \wedge (F(\{e_A\}) = \{e_B\} \wedge \text{Mon}(B, \diamond, e_B))$  | $A$                       |
| 8     | 9  | $e_B \in B$                                                                    | $\wedge E1(8)$            |
| 8     | 10 | $\text{Mon}(B, \diamond, e_B)$                                                 | $\wedge E2(\wedge E2(8))$ |

Jedes Zielelement ist eine Einheit:

|          |    |                                                                                                      |                                       |
|----------|----|------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1        | 11 | $U = A$                                                                                              | Th. 40.3.3.17(1)                      |
| 1,3,8    | 12 | $F(U) = V \wedge F[\mathcal{P}(U) \setminus \{\emptyset\}] = \mathcal{P}(V) \setminus \{\emptyset\}$ | Th. 40.3.3.18(4, 10, 3)               |
| 1,3,8    | 13 | $F[P] = \mathcal{P}(V) \setminus \{\emptyset\}$                                                      | $= E(11, \wedge E2(12))$              |
| 3        | 14 | $F: P \xrightarrow{\sim} Q$                                                                          | Ax. 28.2.12.6(3)                      |
| 3        | 15 | $F[P] = Q$                                                                                           | Th. 8.2.2.1(14)                       |
| 1,3,8    | 16 | $Q = \mathcal{P}(V) \setminus \{\emptyset\}$                                                         | Th. 2.9.1.4(15, 13)                   |
| 17       | 17 | $b \in B$                                                                                            | $A$                                   |
| 17       | 18 | $\{b\} \in Q$                                                                                        | Th. 38.2.3.1(17)                      |
| 1,3,8,17 | 19 | $\{b\} \in \mathcal{P}(V) \setminus \{\emptyset\}$                                                   | $= E(16, 18)$                         |
| 1,3,8,17 | 20 | $\{b\} \subseteq V$                                                                                  | $\wedge E1(\text{Th. 3.16.3.1 (19)})$ |
|          | 21 | $b \in \{b\}$                                                                                        | Th. 3.11.3.8                          |
| 1,3,8,17 | 22 | $b \in V$                                                                                            | Th. 3.5.2.6(20, 21)                   |
| 1,3,8,17 | 23 | $\text{Einh}_{B, \diamond, e_B}(b)$                                                                  | Def. 40.3.3.1(10, 22)                 |
| 1,3,8,17 | 24 | $\exists c \in B (b \diamond c = e_B \wedge c \diamond b = e_B)$                                     | Def. 38.2.5.1(23)                     |
| 1,3,8    | 25 | $\forall b \in B \exists c \in B (b \diamond c = e_B \wedge c \diamond b = e_B)$                     | $\forall I(\rightarrow I(17, 24))$    |
| 1,3,8    | 26 | $\text{Grp}(B, \diamond, e_B)$                                                                       | Def. 40.2.1.1(10, 25)                 |

Gruppenisomorphismus und Existenzschluss:

|          |    |                                                                                                                       |                                          |
|----------|----|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1,3,8    | 27 | $\exists f (\text{Grplso}(f; A, \star, e_A; B, \diamond, e_B) \wedge \forall a \in A (F(\{a\}) = \{f(a)\}))$          | Th. 40.3.3.4(1, 26, 3)                   |
| 28       | 28 | $\text{Grplso}(f; A, \star, e_A; B, \diamond, e_B) \wedge \forall a \in A (F(\{a\}) = \{f(a)\})$                      | $A$                                      |
| 1,3,8,28 | 29 | $\exists f (\text{Grp}(B, \diamond, e_B) \wedge \text{Grplso}(f; A, \star, e_A; B, \diamond, e_B))$                   | $\exists I(\wedge I(26, \wedge E1(28)))$ |
| 1,3,8    | 30 | $\exists f (\text{Grp}(B, \diamond, e_B) \wedge \text{Grplso}(f; A, \star, e_A; B, \diamond, e_B))$                   | $\exists E(27, 28, 29)$                  |
| 1,3,8    | 31 | $\exists e_B \in B \exists f (\text{Grp}(B, \diamond, e_B) \wedge \text{Grplso}(f; A, \star, e_A; B, \diamond, e_B))$ | $\exists I(\wedge I(9, 30))$             |
| 1,2,3    | 32 | $\exists e_B \in B \exists f (\text{Grp}(B, \diamond, e_B) \wedge \text{Grplso}(f; A, \star, e_A; B, \diamond, e_B))$ | $\exists E(7, 8, 31)$                    |

□

*Bemerkung.* Der Beweis verwendet wesentlich die *große* Potenzhalbgruppe aller nichtleeren Teilmengen: Die ganze Einheitengruppe  $A^\times$  muss selbst als Element von  $\mathcal{P}(A) \setminus \{\emptyset\}$  auftreten. Für endliche Grundträger stimmt die große mit der finitären Potenzhalbgruppe überein. Bei unendlichen Gruppen folgt die entsprechende Aussage für die Halbgruppe nur der endlichen nichtleeren Teilmengen dagegen nicht aus diesem Beweis.